DoMS

QUIET EXCELLENCE DEMONSTRATED

MS3520 : Operations Management I

INVENTORY CONTROL

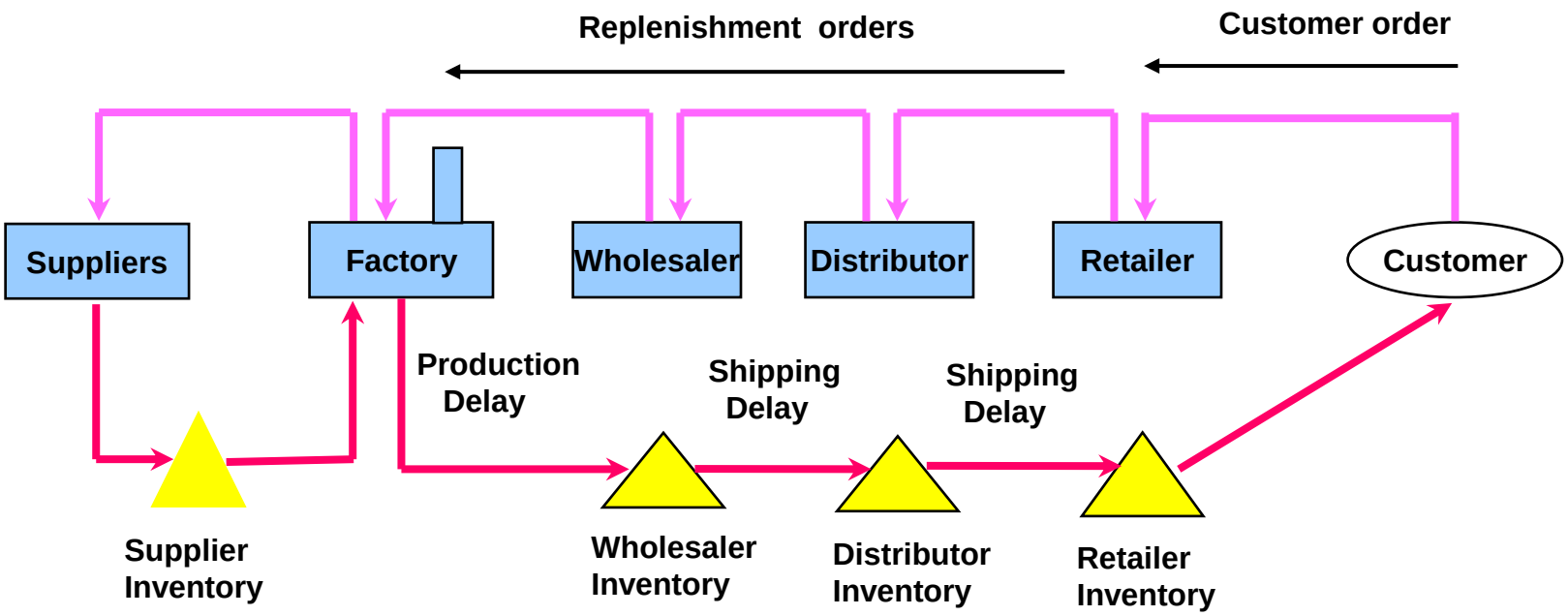
Inventory

The stock of any item or resource used in an organization.

These items or resources can include:

- Raw Materials
- WIP
- Finished Goods
- Pipeline inventory
- Machine Spares

Why Inventory Control

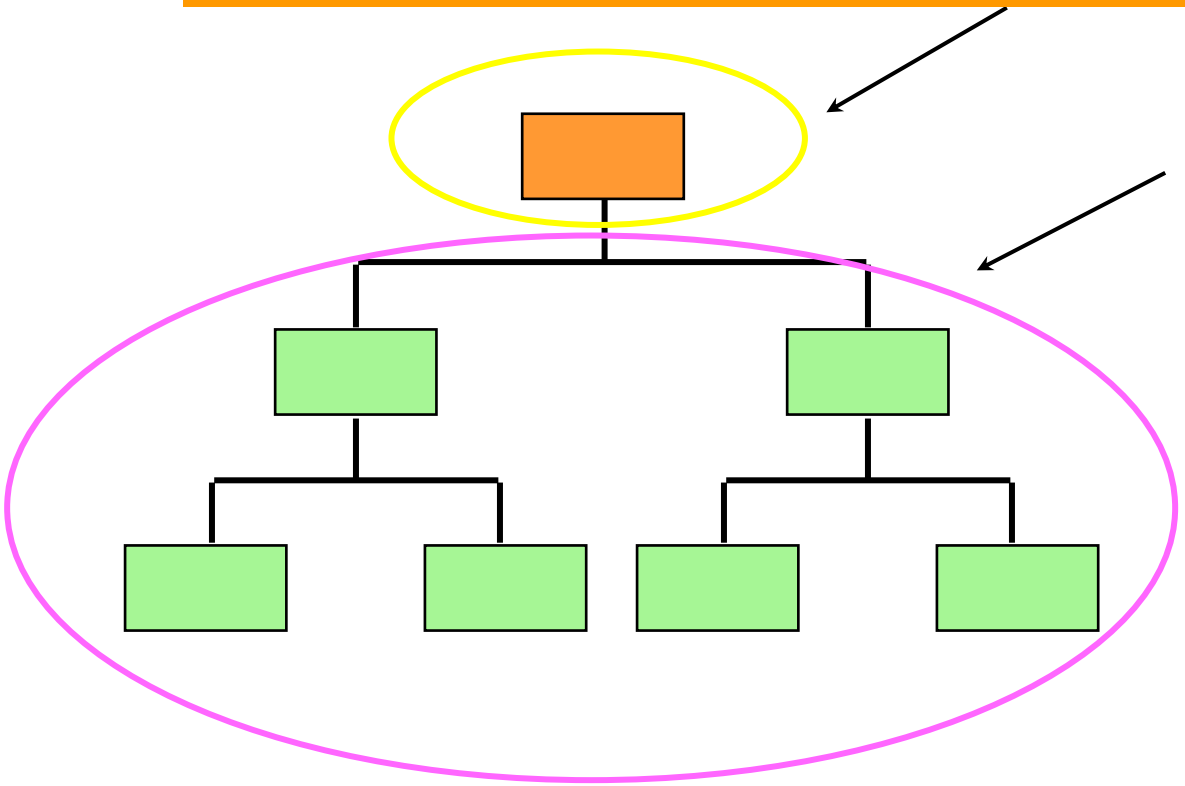


Purposes of Inventory

1. To maintain independence of operations
2. To meet variation in product demand
3. To allow flexibility in production scheduling
4. To provide a safeguard for variation in raw material delivery time
5. To take advantage of economic purchase-order size

Independent vs. Dependent Demand

Independent Demand (Demand not related to other items or the final end-product)



Dependent Demand (Derived demand items for component parts, subassemblies, raw materials, etc.)

Costs in Inventory Control

- Item cost or unit cost
- Ordering / set-up cost
  - Clerical
  - Administrative
  - Expediting
  - Transportation etc.

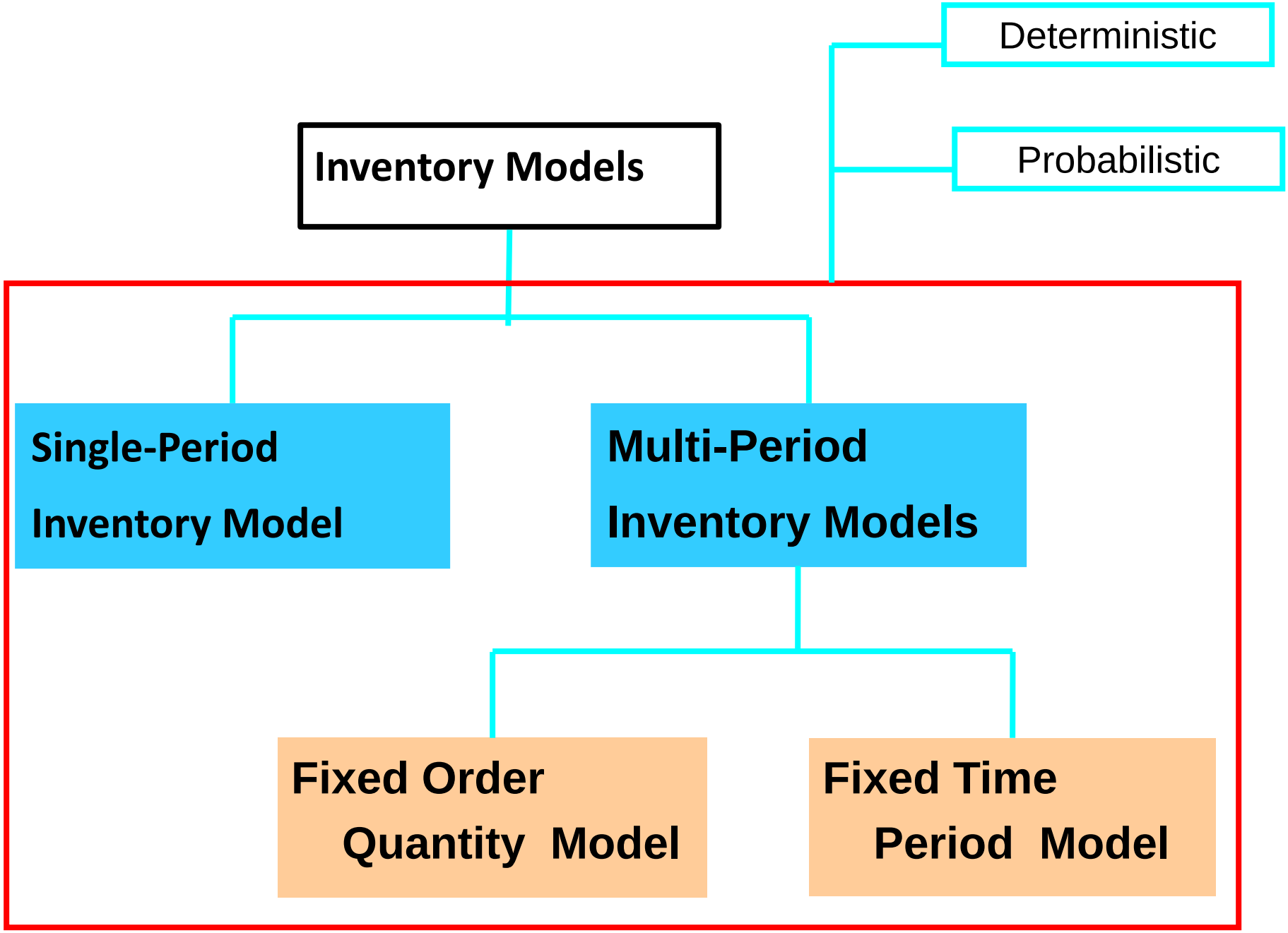
Costs in Inventory Control

- Shortage costs
  - Overtime requirements due to shortage
  - Special clerical and administrative costs, expediting
  - Loss of goodwill
  - Special transportation
  - Lost production time

Costs in Inventory Control

- Inventory carrying costs
  - Locked-up capital
  - Storage space
  - Handling
  - Taxes
  - Insurance
  - Obsolescence
  - Deterioration
  - Maintaining records

Inventory Models


Deterministic
Probabilistic
Inventory Models
Single-Period Inventory Model
Multi-Period Inventory Models
Fixed Order Quantity Model
Fixed Time Period Model

Single Period Inventory Model

Single - Period Probabilistic Inventory Model

$$P \leq \frac{Cu}{Co + Cu}$$

This model states that we should continue to increase the size of the inventory so long as the probability of selling the last unit added is equal to or less than the ratio of: Cu/(Co+Cu)

Newsvendor(/boy) Problem

- Often managers have to make decisions about inventory level over a very limited period.
- For a newsvendor who sells papers on a street corner, the demand is uncertain.
- The newsvendor must decide how many papers to buy from his supplier.
- If he buys too many papers he is left with unsold papers that have no value at the end of the day;
- If he buys too few papers he has lost the opportunity of making a higher profit.

Newsvendor model - Properties

- One-time decision
  - Current decisions only impact the "next period" but not future periods
    - Retail: fashion/seasonal items
    - One-time events
- Relevant costs
  - C_o: Unit cost of excess inventory (cost of overage)
  - C_u: Unit cost of shortage (cost of underage)
- Demand D is a random variable with
  - Probability density function (pdf), $f(x)$
  - cumulative distribution function (cdf), $F(x)$
- Objective: Choose the ordering quantity Q (before you know the demand) to minimize
  - Total **expected** overage and underage costs

Newsvendor model – Finding the optimal order quantity

- First, write the cost function:

$$G(Q,D): \text{total overage} + \text{underage cost} \quad (\text{if } Q \text{ is ordered and demand is } D)$$

$$G(Q,D) := \begin{cases} c_u(D-Q) & \text{if} \quad D \geq Q \\ c_o(Q-D) & \text{if} \quad D \leq Q \end{cases}$$

$$G(Q) = \text{expected} \quad \text{overage} + \text{underage cost} \ \text{ if } Q \text{ is ordered}$$

$$G(Q) = \mathrm{E}[G(Q,D] = \int_0^\infty G(Q,x) f(x) dx$$

Critical Ratio

$$= \int_0^Q c_o(Q-D) f(x) dx + \int_Q^\infty c_u(D-Q) f(x) dx \Rightarrow F(Q) = \boxed{\frac{c_u}{c_u + c_o}}$$

Derivation of the Critical Ratio - 1

$$G(Q) = \text{expected overage} + \text{underage cost if } Q \text{ is ordered}$$

$$G(Q) = c_o \int_0^Q (Q-D) f(x)dx + c_u \int_Q^\infty (D-Q) f(x)dx = c_o G_1 + c_u G_2$$

To take the derivative of G(Q), use Leibniz's Rule

$$\frac{d}{dy} \int_{f_1(y)}^{f_2(y)} h(x,y)dx = \int_{f_1(y)}^{f_2(y)} \frac{\partial h(x,y)}{\partial y} dx + h(f_2(y), y).f_2'(y) - h(f_1(y), y).f_1'(y)$$

In G1: $y \leftrightarrow Q \qquad f_2(y) \leftrightarrow Q \qquad f_1(y) \leftrightarrow 0$

$$h(x,Q) = (Q-x) f(x)$$

Derivation of the Critical Ratio - 2

$$G_1(Q) = \int_0^Q (Q-D) f(x)dx$$

In G1: $y \leftrightarrow Q$ $\quad f_2(y) \leftrightarrow Q$ $\quad f_1(y) \leftrightarrow 0$

$$h(x,Q) = (Q-x) f(x)$$

To take the derivative of G(Q), use Leibniz's Rule

$$\frac{d}{dy}\int_{f_1(y)}^{f_2(y)} h(x,y)dx = \int_{f_1(y)}^{f_2(y)} \frac{\partial h(x,y)}{\partial y} dx + h(f_2(y),y).f_2'(y) - h(f_1(y),y).f_1'(y)$$

$$\frac{\partial h(x,y)}{\partial y} = \frac{\partial[(Q-D)f(x)]}{\partial Q} = f(x)$$

$$h(f_2(y),y) = h(Q,Q) = (Q-Q)f(x) = 0 \qquad f_2'(y) = 1$$

$$h(f_1(y),y) = h(0,Q) = (Q-0)f(x) = Qf(x) \qquad f_1'(y) = 0$$

$$\Rightarrow \frac{d}{dQ}G_1(Q) = \int_0^Q f(x)dx \quad \text{Similarly,} \quad \frac{d}{dQ}G_2(Q) = -\int_Q^\infty f(x)dx$$

Derivation of the Critical Ratio - 3

$$\frac{d}{dQ}G(Q) = c_o\int_0^Q f(x)dx - c_u\int_Q^\infty f(x)dx = c_oF(Q) - c_u(1-F(Q))$$

$$= (c_o + c_u)F(Q) - c_u = 0$$

$$\Rightarrow F(Q) = \frac{c_u}{c_u + c_o}$$

Recall: $\int_0^\infty f(x)dx = 1$

$$\int_0^Q f(x)dx = F(Q) = P(D \le Q) \qquad \int_Q^\infty f(x)dx = 1 - F(Q) = P(D \ge Q)$$

Also need to check if G(Q) is convex. Second derivative is $(c_o+c_u)f(Q)\ge 0$

Example

- University is planning to order annual day T - shirts for sales this weekend.
- Based on our past experience we sell on average 2,400 T - shirts with a standard deviation of 350.
- We make ₹100 on every T - shirt we sell at the stall, but lose ₹10 on every T - shirt not sold.
- How many shirts should we order ?

C_u = ₹100 and C_o = ₹10;

$P \leq 100 / (10 + 100) = 0.909$

Assuming demand to be normally distributed,

$Z_{.909}$ = 1.335 (NORM.S.INV(.909))

therefore we need 2,400 + 350*(1.335) = 2868 shirts

Multi period Inventory Model

Multi-Period Inventory Models

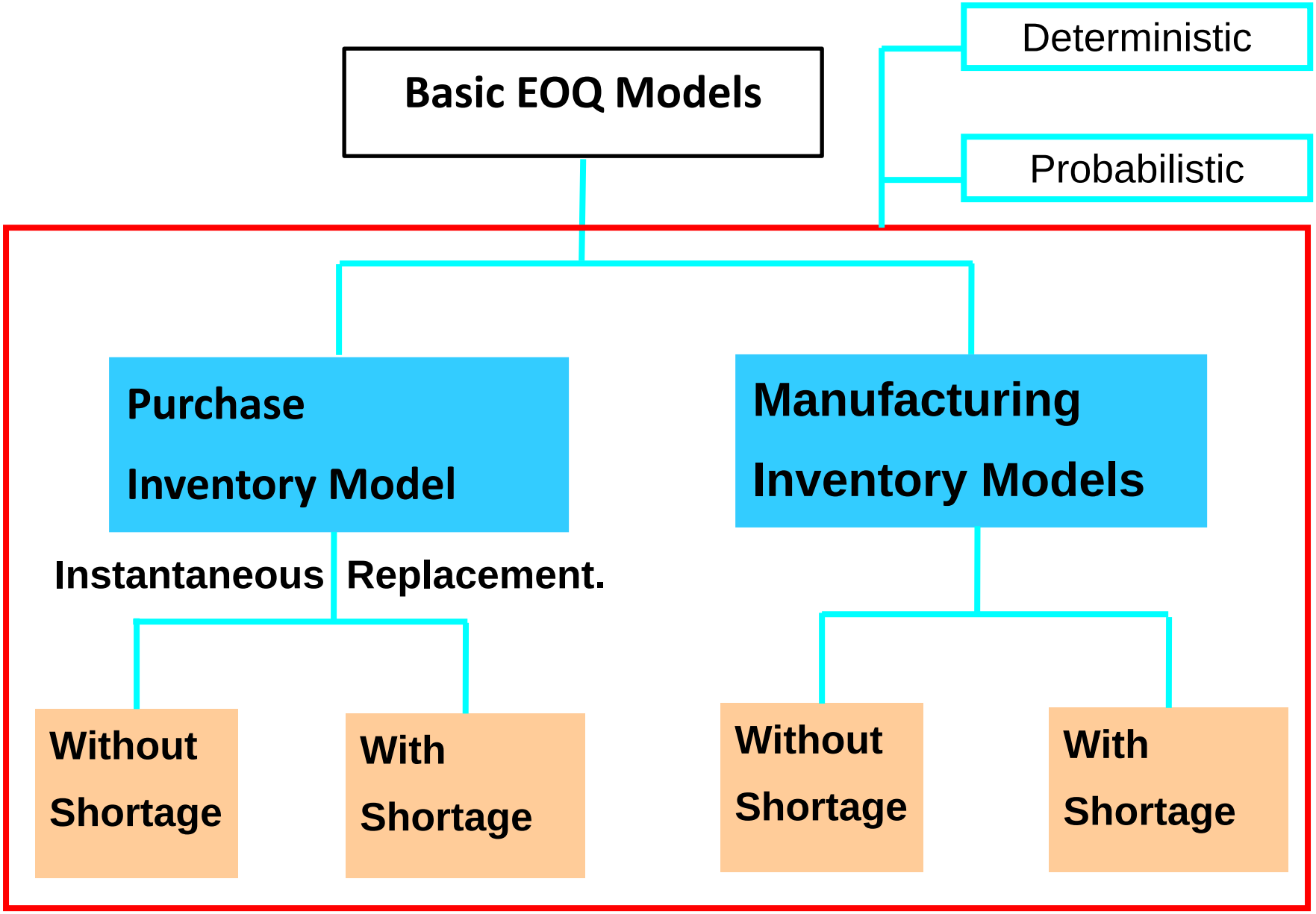
- In a single-period inventory model, inventory decisions are made only for one future period.

- However, in a multi-period inventory model, inventory decisions are made in the present for several future periods.

- In such models, the objective is to determine the optimal ordering quantity as well as the optimal ordering point.

Multi-Period Inventory Models

- **Fixed-Order Quantity Models (Continuous Review / "Q Model")**
  - Order quantity is fixed (EOQ), but time between orders is variable
  - Event triggered (Example: running out of stock or reaching a predetermined reorder level)
  - Records must be perpetually maintained
- **Fixed-Time Period Models (Periodic Review / "P Model")**
  - Inventory is examined only at periodic intervals
  - Time triggered (Example: Monthly sales call by sales representative)
  - Higher average inventory to protect against stock-out during review period

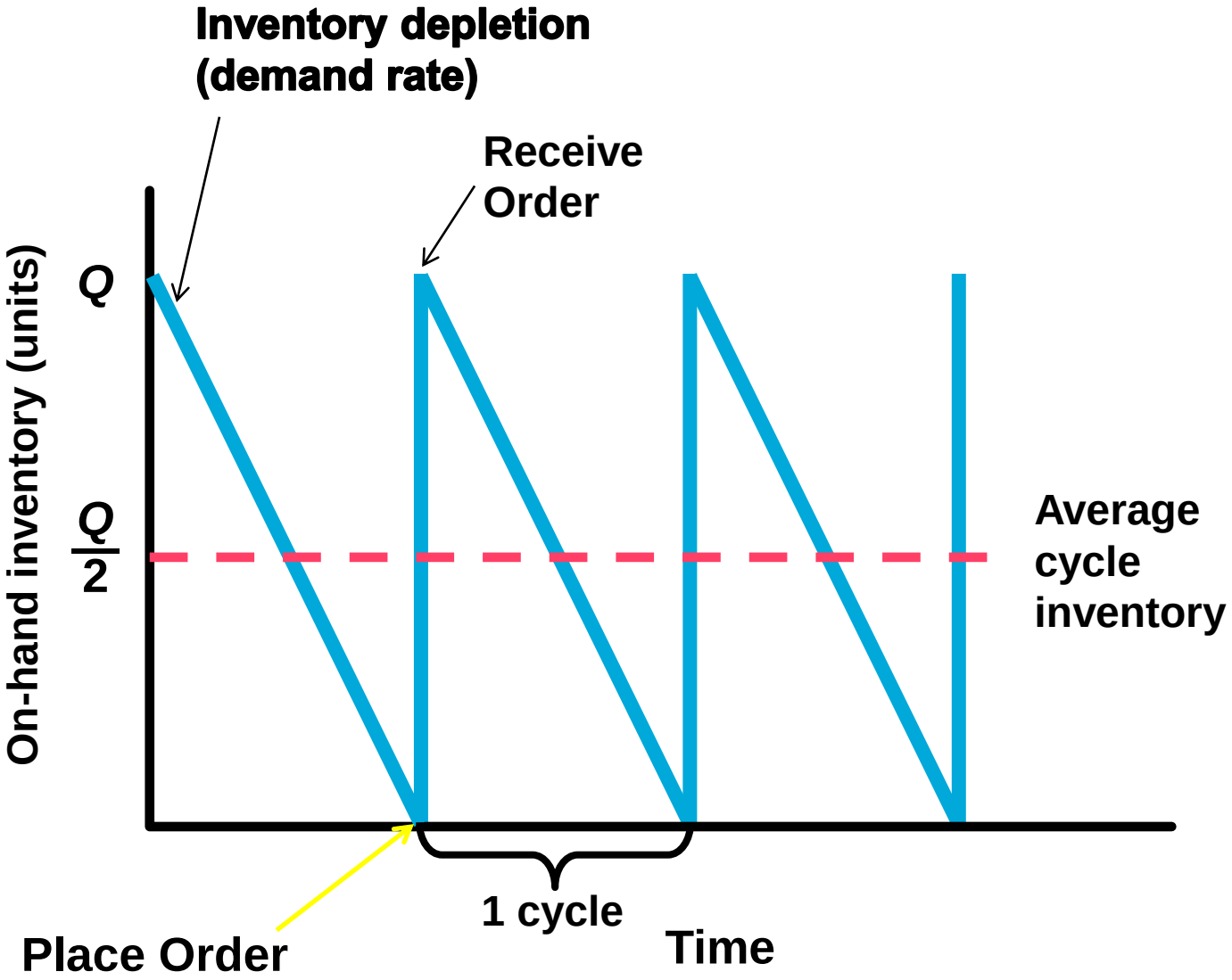

Basic EOQ Models
Deterministic
Probabilistic
Purchase Inventory Model
Manufacturing Inventory Models
Instantaneous Replacement.
Without Shortage
With Shortage
Without Shortage
With Shortage

Deterministic EOQ models

Assumptions

- Demand rate is constant
- No constraints on lot size
- Only relevant costs are holding and ordering/setup costs
- Unit cost of an item is constant
- No uncertainty in lead time or supply

Inventory model without shortage



Notations

- D = Annual Demand in units per year
- Cu = Material Cost in Rs per unit
- C_c = Carrying Cost or Holding Cost in Rs per unit per time
- Co = Ordering Cost or Setup Cost in Rs per lot
- Q = EOQ in units per lot

EOQ Formula Derivation

D = Annual demand (units)
C_u = Cost per unit (₹)
Q = Order quantity (units)
C_o = Cost per order (₹)
I = Ratio of Holding cost per unit per time to unit price(%)
C_C = Holding cost per unit per time (₹) = $I*C_u$

Number of Orders = D / Q
Ordering costs = C_o x (D / Q)

Average inventory in terms of
- **units = Q / 2**
- **Rs = (Q / 2) x C_C**

Cost to carry average inventory = (Q /2) x C_c
= (Q / 2) x I x C_U

Total cost = *inv carry cost* + *order cost*
= (Q/2) x C_C + C_O x (D/Q)

Take the 1st derivative:

$d(TC)/d(Q) = (C_C) / 2 - (D \times C_O) / Q^2$

To optimize: set $d(TC)/d(Q) = 0$

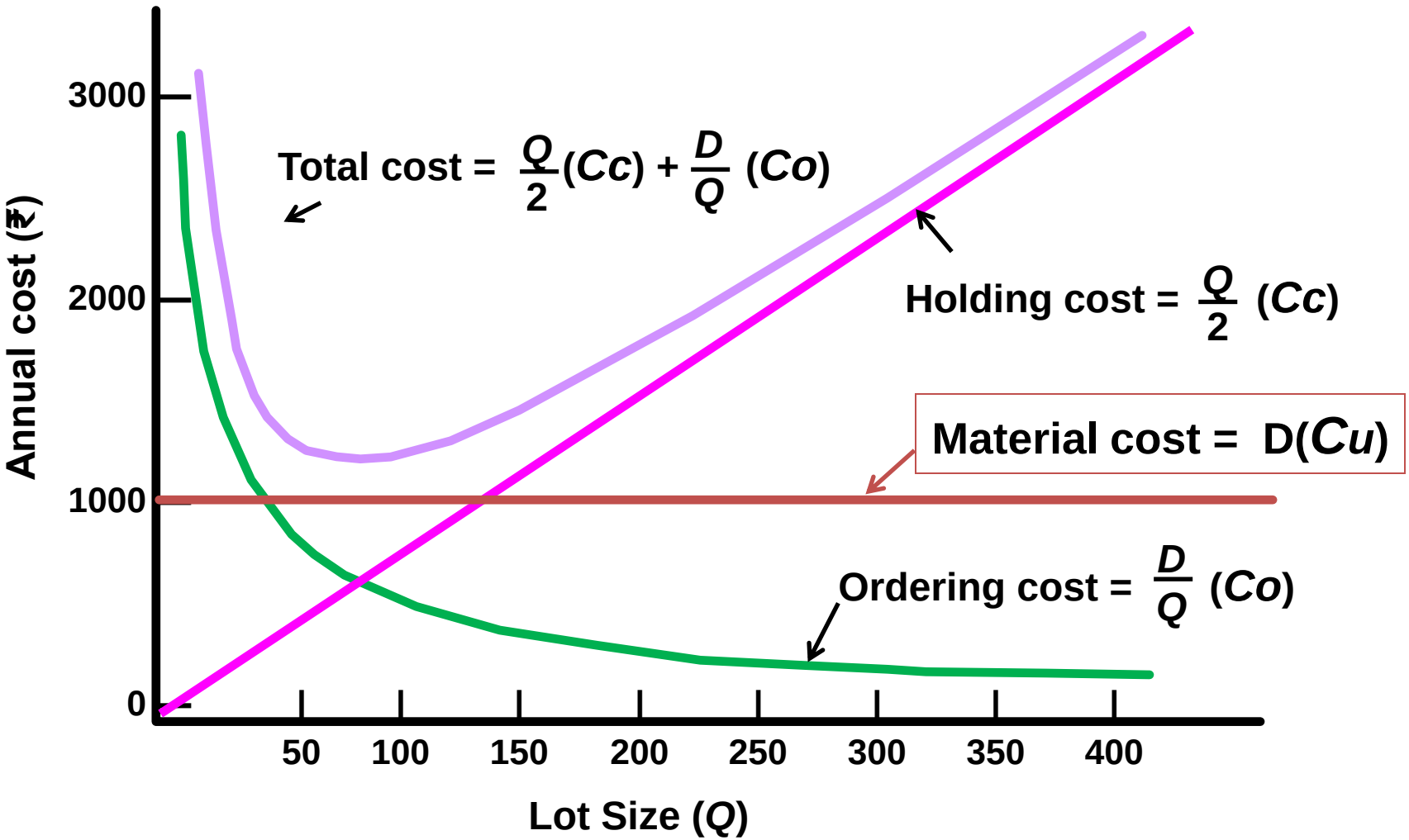
$DC_o / Q^2 = C_C / 2$

$Q^2/DC_o = 2 / C_C$

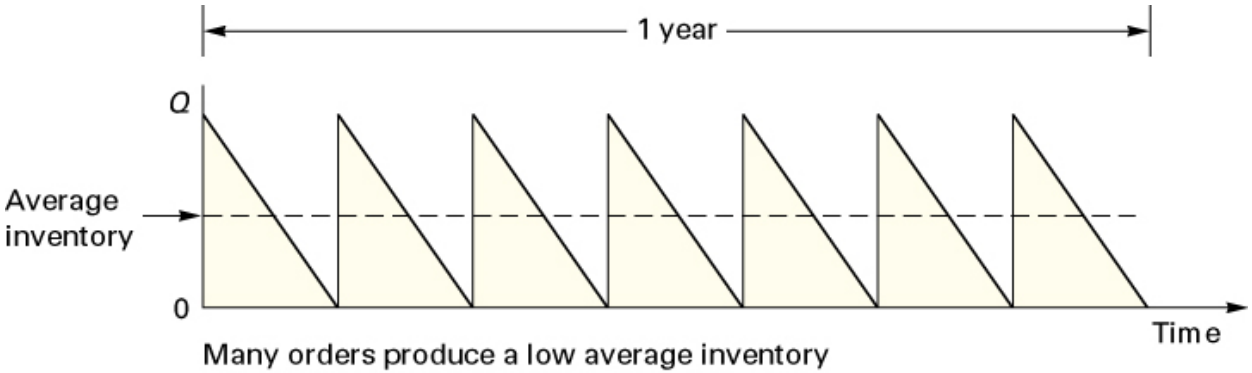
$Q^2 = (DC_O \times 2) / C_C$

$$Q^* = \sqrt{\frac{2DC_o}{C_c}}$$

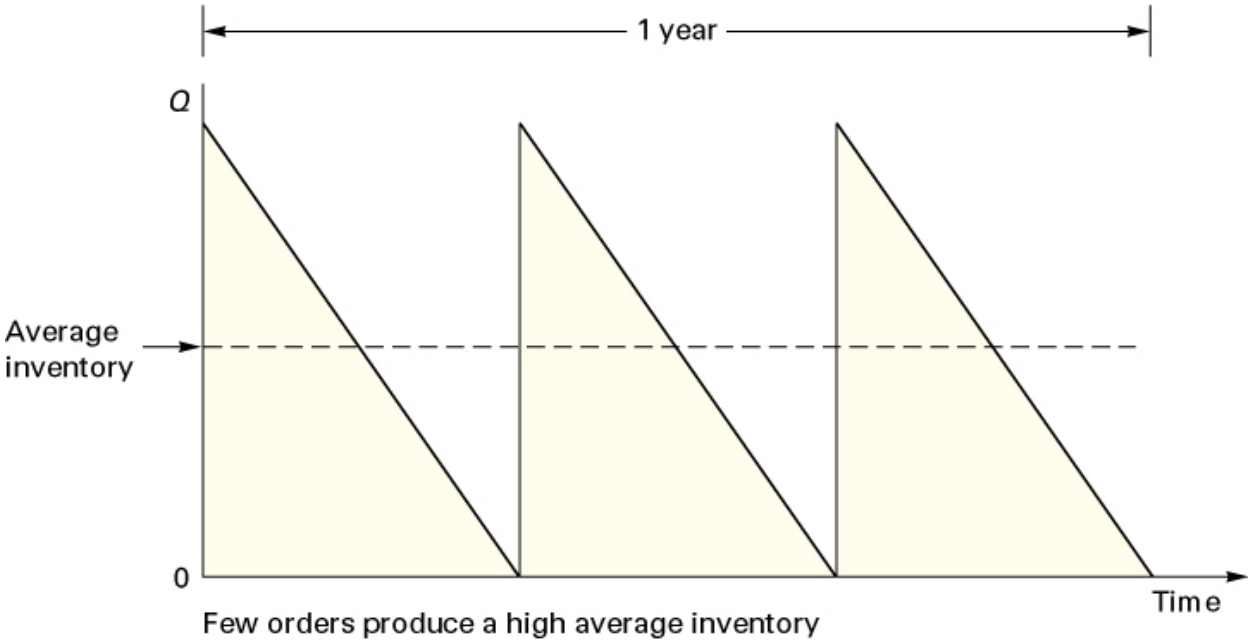
Economic Order Quantity



Effect of Order frequency



Many orders produce a low average inventory

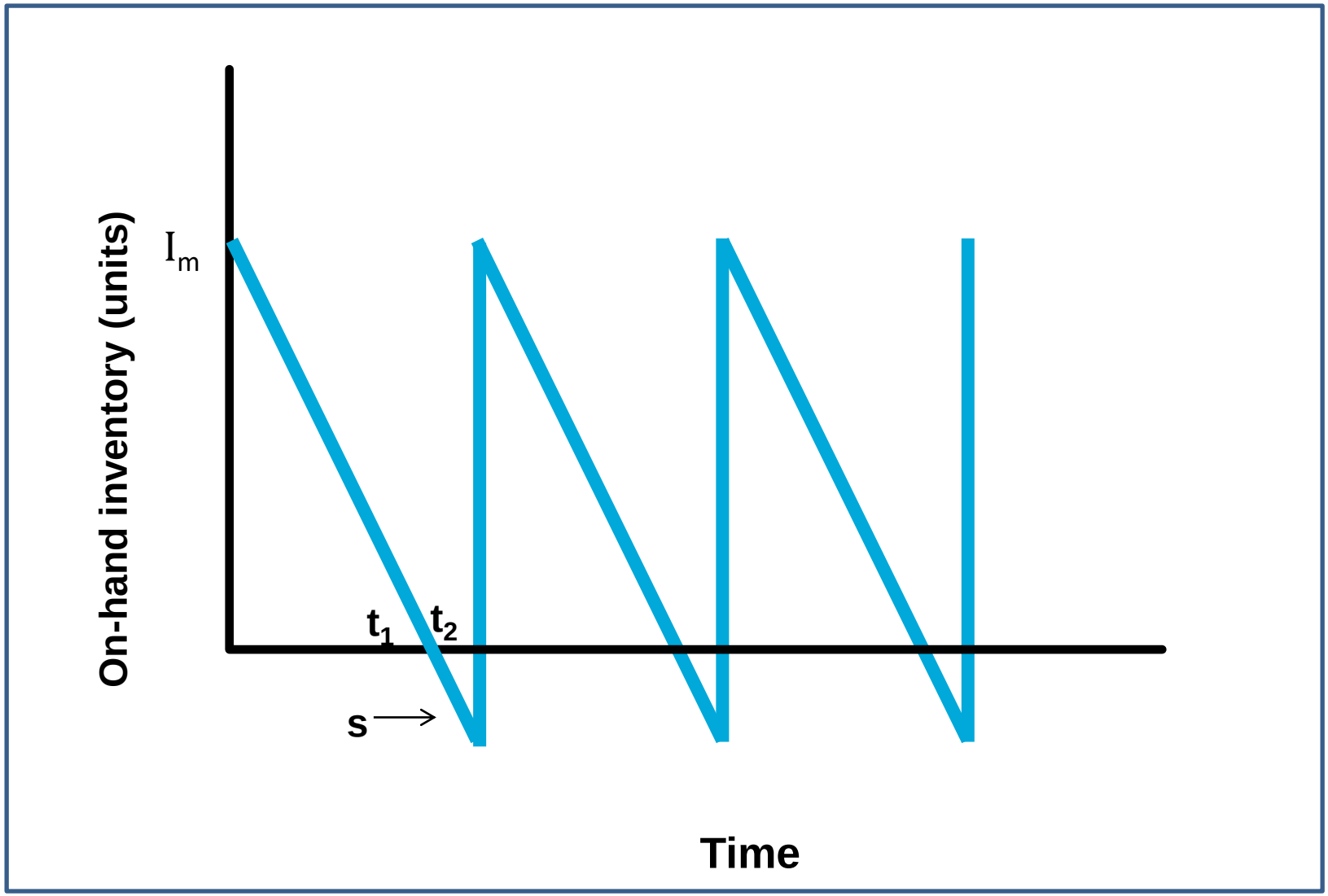


Few orders produce a high average inventory

No of orders per year

= D / Q

Inventory model with shortage



Notations and EOQ

D = Annual Demand in units per year

Cu = Material Cost in ₹ per unit

C_C = Carrying Cost in ₹ per unit per time

Co = Ordering Cost or Setup Cost in ₹ per lot

Cs = Shortage Cost in ₹ per unit per time

s = Shortage quantity

Q = EOQ in units per lot

I_m = Maximum inventory in a cycle (Q-s)

t_1= inventory period per cycle

t_2= shortage period per cycle

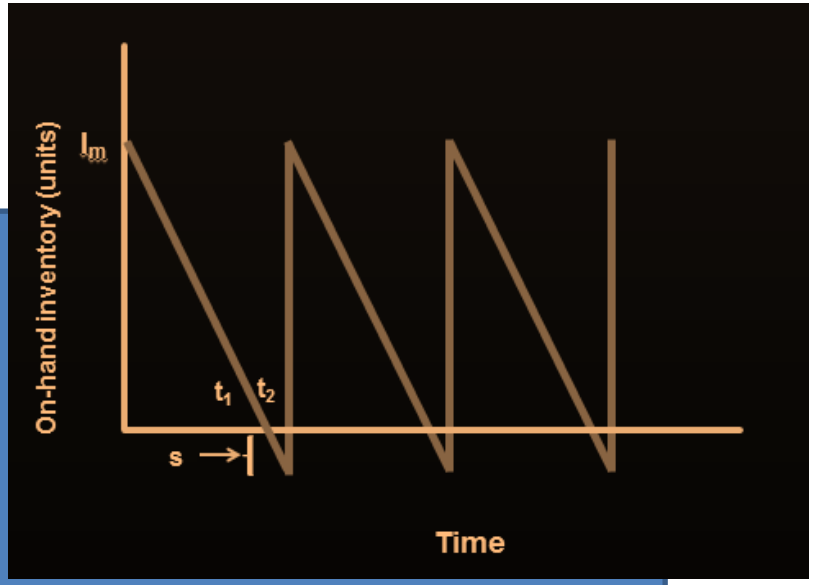
$$Inventory\ carry\ cost = \frac{I_m}{2} \times C_c \times \frac{t_1}{(t_1 + t_2)}$$

$$order\ cost = \frac{D}{Q} \times C_0$$

$$shortage\ cost = \frac{s}{2} \times C_s \times \frac{t_2}{(t_1 + t_2)}$$

$$Total\ cost = \frac{s}{2} \times C_s \times \frac{t_1}{(t_1 + t_2)} + \frac{D}{Q} \times C_0 + \frac{I_m}{2} \times C_c \times \frac{t_2}{(t_1 + t_2)}$$

- From similar triangles, we get



$$\frac{t_1}{(t_1+t_2)} = \frac{(Q-s)}{Q}$$

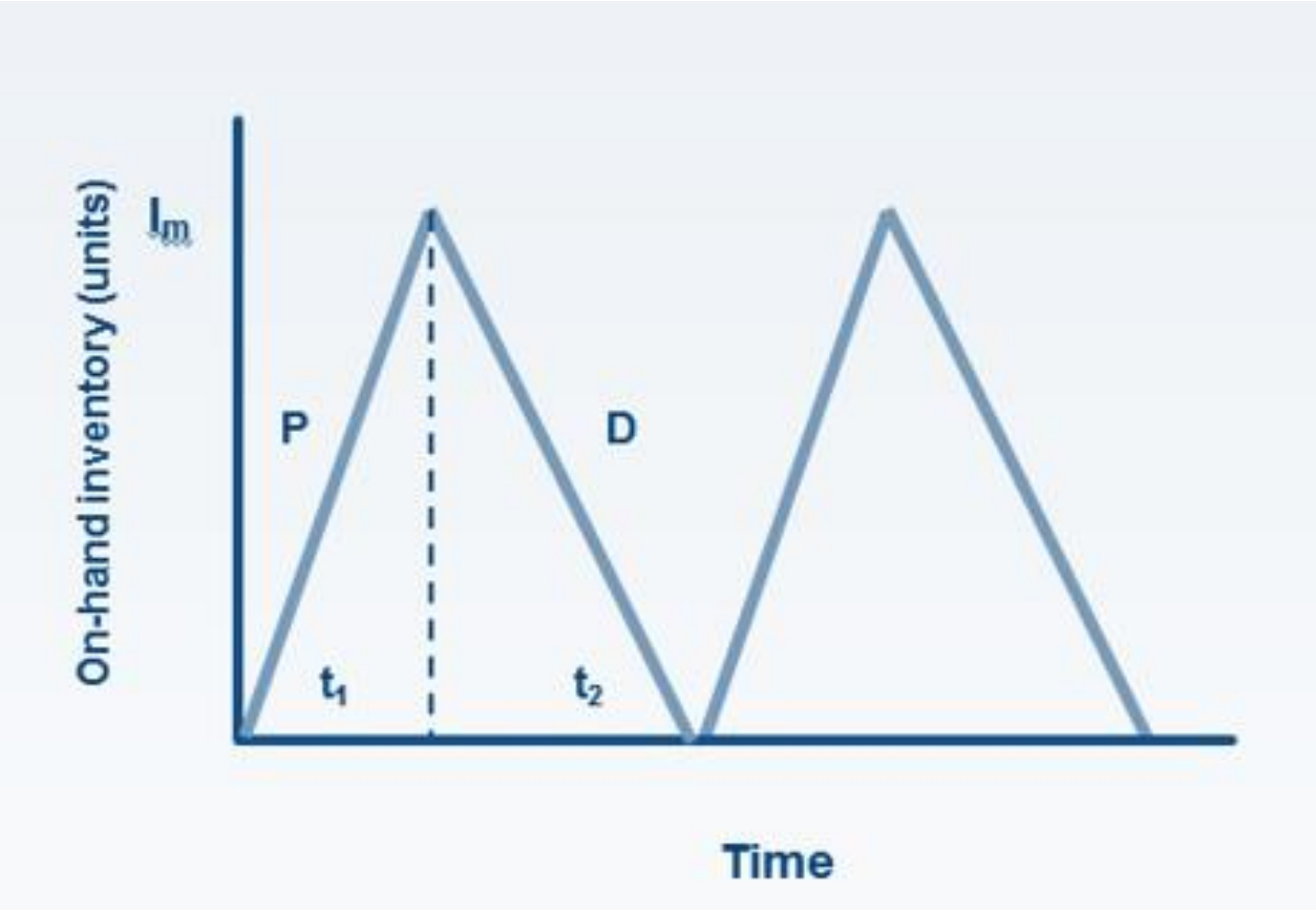
$$\frac{t_2}{(t_1+t_2)} = \frac{s}{Q}$$

$$TC = \frac{DC_0}{Q} + \frac{(Q-s)^2 C_c}{2Q} + \frac{s^2 C_s}{2Q}$$

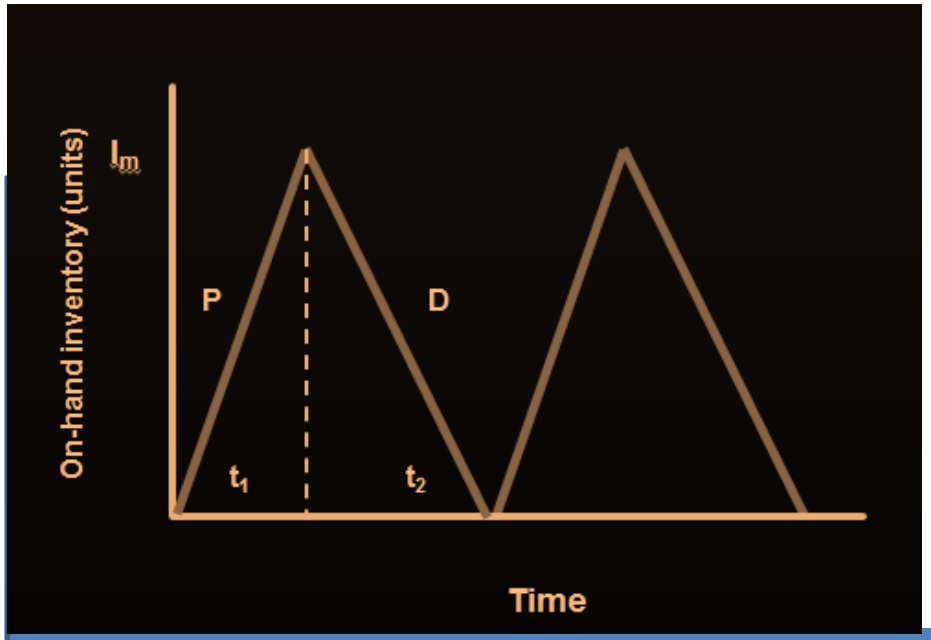
Partial derivative *w.r.t Q* and *s* and setting to 0

$$\mathbf{Q^* = \sqrt{\frac{2DC_o(C_c + C_s)}{C_c C_s}}} \qquad \mathbf{s = \frac{Q \times C_c}{(C_c + C_s)}}$$

Production consumption model



Notations and EOQ



D = Annual Demand in units per year

Cu = Material Cost in ₹ per unit

C_C = Carrying Cost in ₹ per unit per time

Co = Ordering Cost or Setup Cost in ₹ per lot

Q = EOQ in units per lot

I_m = Maximum inventory in a cycle

P = Production rate per year (P>D)

t_1= production period

t_2= consumption period

Inventory carry cost = $(I_m/2) \times C_C$

order cost = $C_O \times (D/Q)$

$$TC = \frac{DC_0}{Q} + \frac{I_m C_c}{2}$$

Substituting Q= P t_1 ; I_m = (P-D) t_1

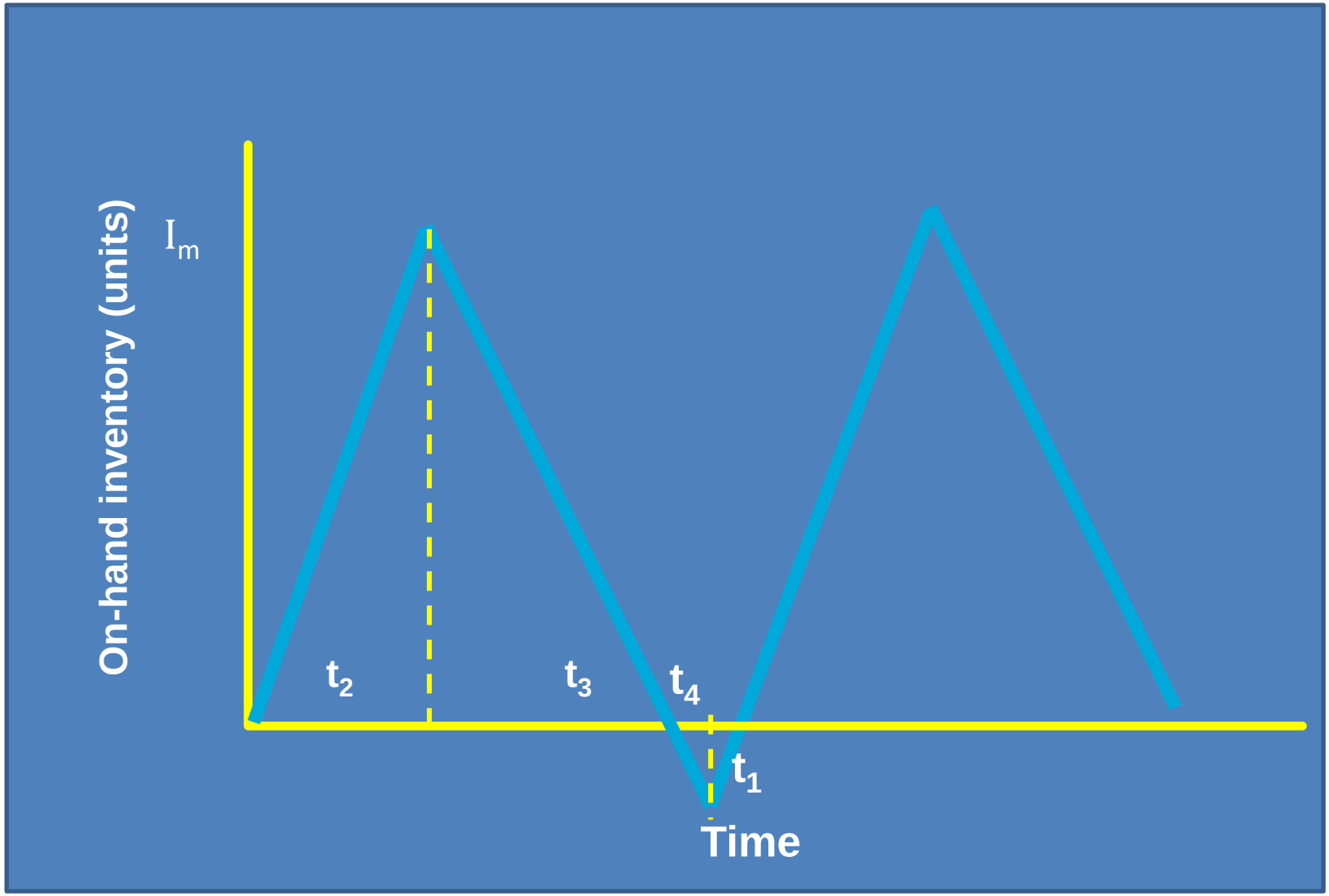
$$TC = \frac{DC_0}{Q} + \frac{Q\left(1-\frac{D}{P}\right)C_c}{2}$$

Partial derivative wrt *Q* and setting to 0

$$-\frac{DC_0}{Q^2} + \frac{\left(1-\frac{D}{P}\right)C_c}{2} = 0$$

$$\mathbf{Q^* = \sqrt{\frac{2DC_o}{C_c\left(1-\frac{D}{P}\right)}}}$$

Production consumption model with shortage



EOQ

Annual Demand = D/year
Production rate = P/year
Order cost per order = C_o
Carrying cost per unit per time = C_c
Shortage cost per unit per time = C_s
Order quantity = Q
Back order allowed = *S*
Maximum inventory in a cycle = I_m
Number of orders/year = $\frac{D}{Q}$
Annual order cost = $\frac{D\ C_o}{Q}$
Average inventory in the system = $\frac{I_m}{2}$
Annual inventory carrying cost = $\frac{I_m C_c}{2} \times \frac{(t_2+t_3)}{T}$
Average quantity back ordered = $\frac{S}{2}$
This is held for a period $(t_1 + t_4)$ out of a cycle of *T.*
Annual back order cost = $\frac{S.C_s}{2} \times \frac{(t_1+t_4)}{T}$

EOQ

TC = $\frac{D\ C_o}{Q} + \frac{I_m C_c}{2} \times \frac{(t_2+t_3)}{T} + \frac{S.C_s}{2} \times \frac{(t_1+t_4)}{T}$

Q = P $(t_1 + t_4)$

$I_m = (P - D)t_2 = Dt_3$

$S = (P - D)t_1 = Dt_4$

$T = t_1 + t_2 + t_3 + t_4$

Q = DT

$\frac{I_m}{Q} = \frac{(P-D)}{P}\ X\ \frac{t_2}{t_1+t_2}$

$$\frac{Q(1 - D/P)}{I_m} = 1 + \frac{t_1}{t_2}$$

From similar triangles $\frac{t_1}{t_2} = \frac{I_m}{S}$

Substituting we get $I_m = Q(1 - D/P) - S$

Also

$$\frac{Pt_2}{T} = \frac{D(t_2+t_3)}{T}$$

$$\frac{(t_2+t_3)}{T} = \frac{Pt_2}{DT} = \frac{PI_m}{Q(P-D)}$$

And

$$\frac{(t_1+t_4)}{T} = \frac{S}{Q(1-D/P)}$$

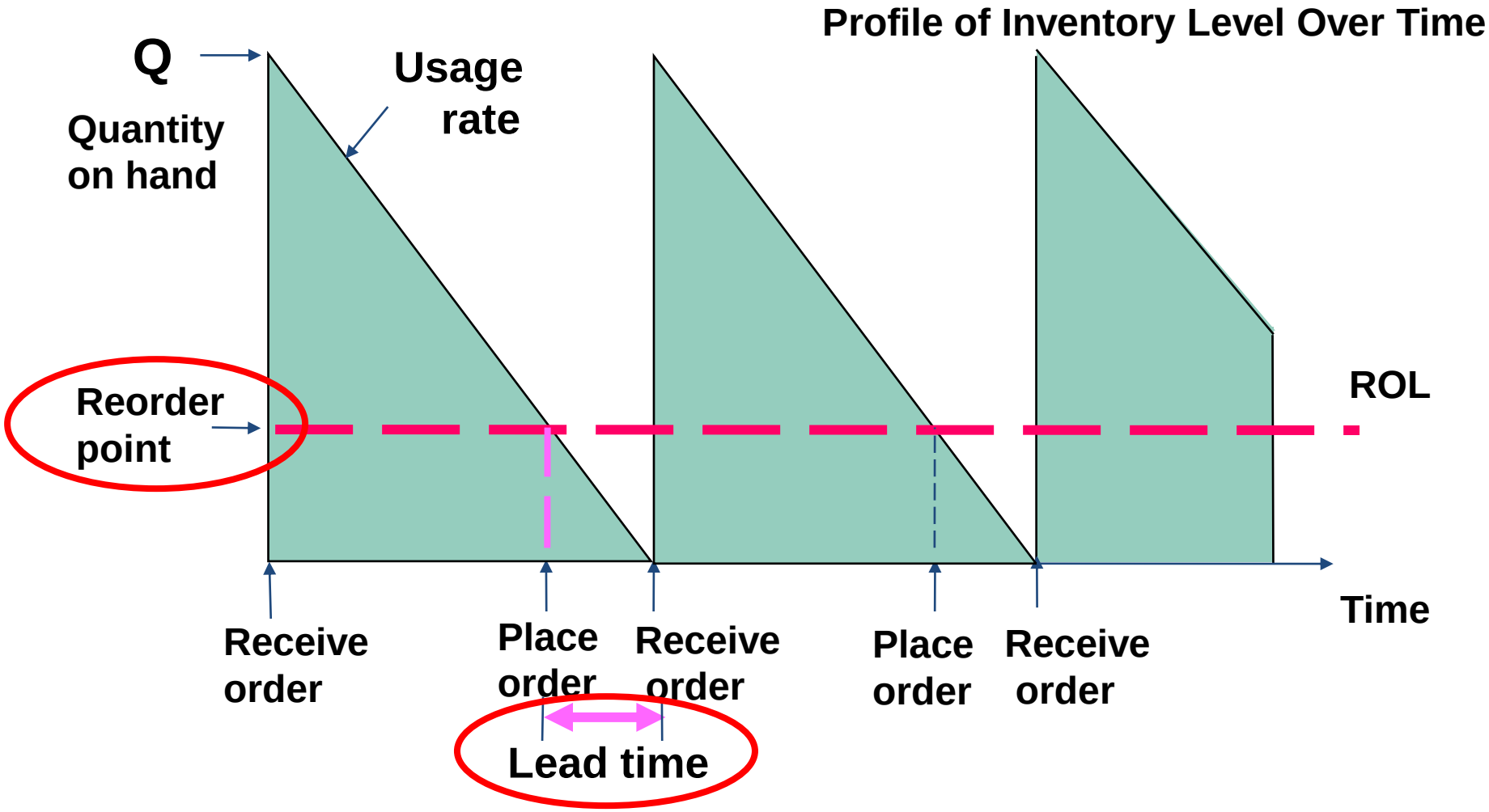
Total cost TC = $\frac{D\,C_o}{Q} + [Q(1-D/P)-S]^2 \times \frac{C_c}{2} \times \frac{1}{Q(1-D/P)} + \frac{S^2.C_s}{2Q(1-D/P)}$

The values of Q and S that minimize the total cost are obtained by setting the first derivative with respect to Q and S to zero.

$$Q^* = \sqrt{\frac{2DC_0(C_c+C_s)}{C_c(1-D/P)C_s}}$$

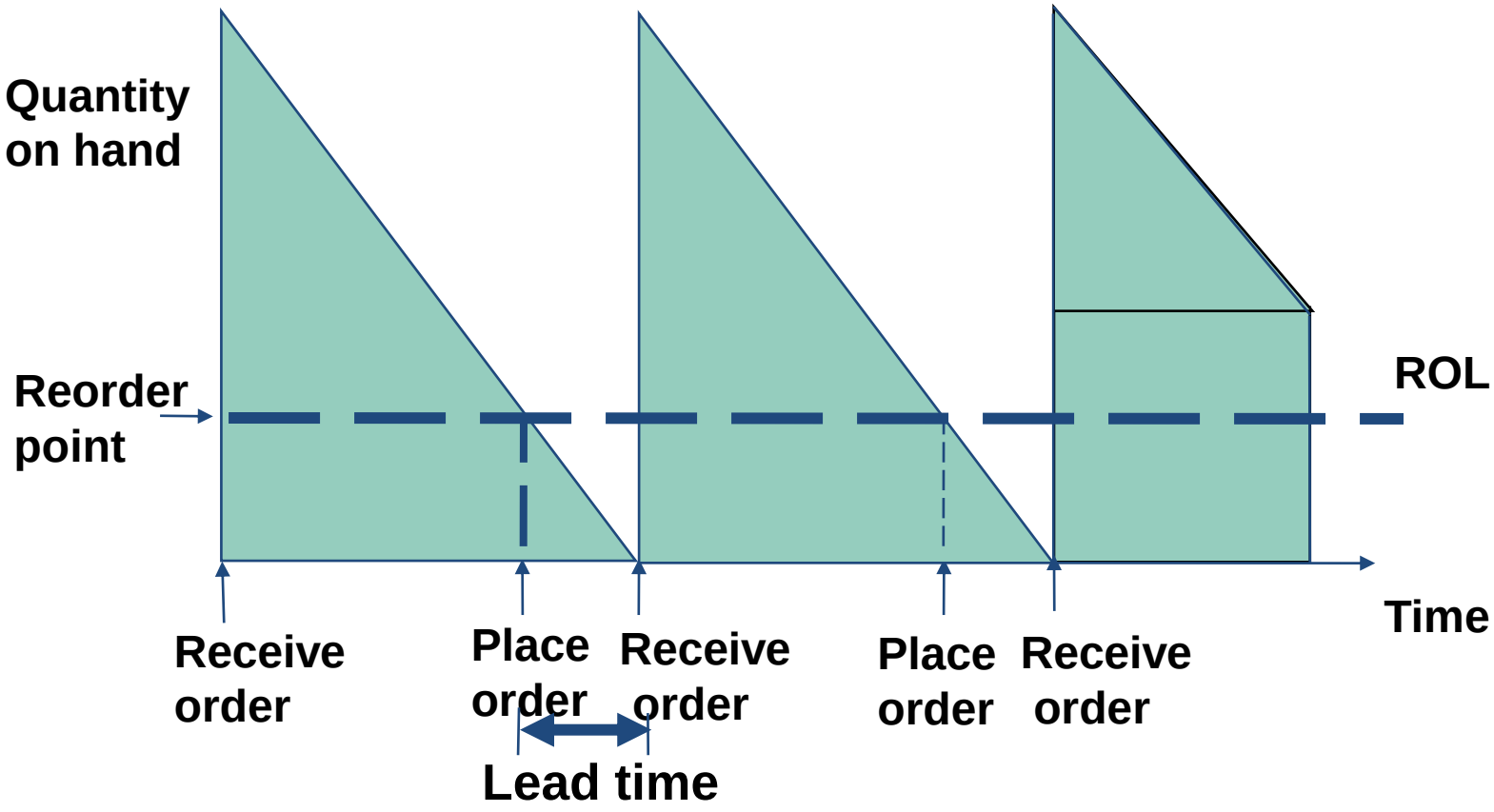
$$S = \frac{QC_c(1-D/P)}{(C_c+C_s)}$$

Effect of Lead Time



Determination of ROL

- ROL = Lead Time Demand
  = Average daily demand x Lead Time in days
  = $\bar{d}$ (LT)



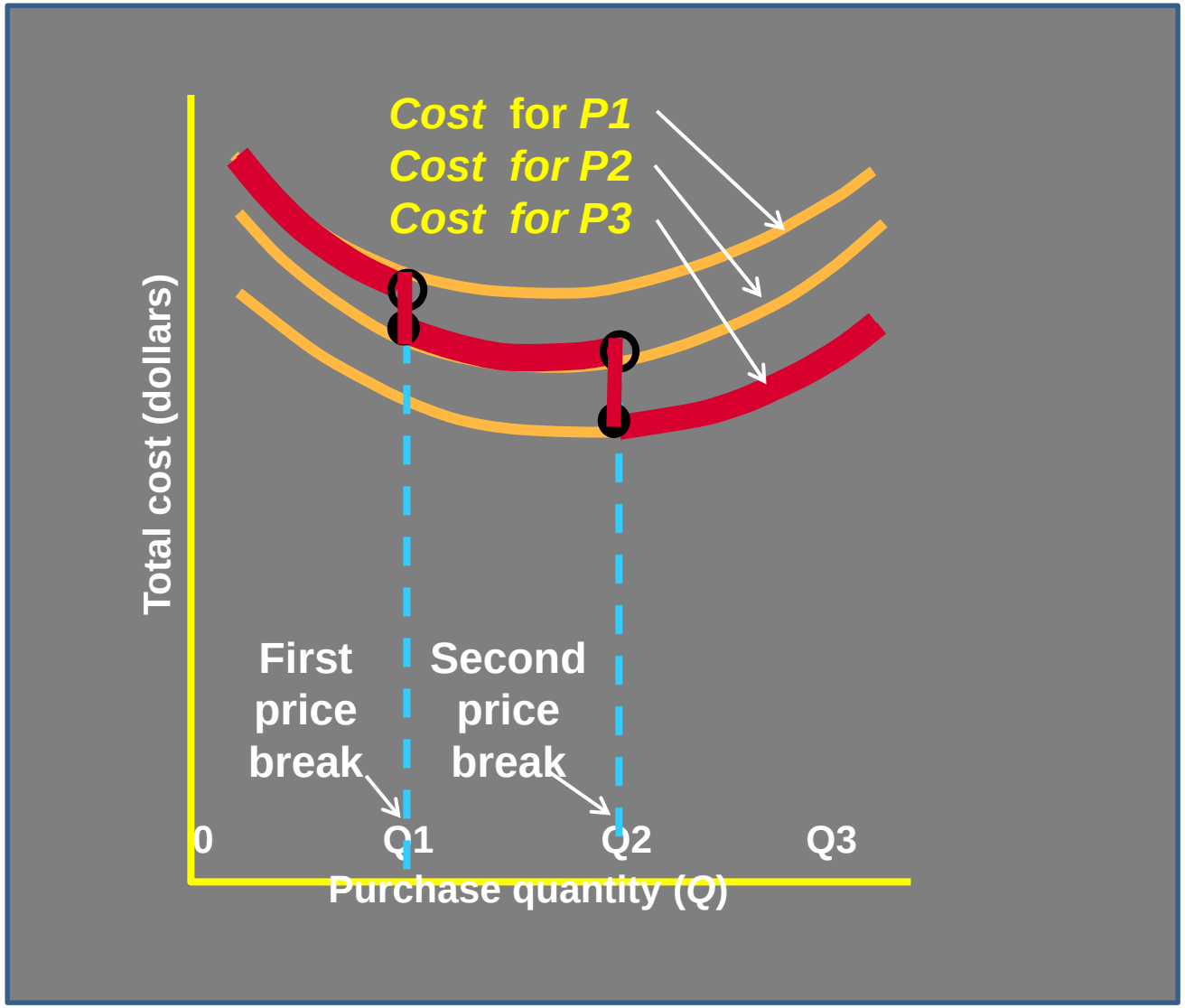
Discount Quantity Models

- Reduced prices are often available when larger quantities are purchased
- Trade-off is between reduced product cost and increased holding cost

Total Cost = Setup cost + Holding Cost + Product cost

Steps to Analyse Discount Models

1. Calculate Q* for each discounted unit price (or start with calculating Q* with minimum unit price).
2. If Q* is too small to qualify for that unit price, adjust Q* upwards
3. Calculate total cost for each Q*
4. Select the Q* with the lowest total cost


Cost for P1
Cost for P2
Cost for P3
Total cost (dollars)
First price break
Second price break
0
Q1
Q2
Q3
Purchase quantity (Q)

Order Quantity	Price per Unit
0 – 299	₹60.00
300 – 499	₹58.80
500 or more	₹57.00

Annual demand = 936 units
Ordering cost = ₹45
Holding cost = 25% of unit price

$$EOQ_{57.00} = \sqrt{\frac{2(936)(45)}{0.25(57.00)}} = 77 \text{ units}$$

Keeping the unit price ₹57, Q* does not lie in the permissible quantity range (500 or more)

$$EOQ_{58.80} = \sqrt{\frac{2(936)(45)}{0.25(58.80)}} = 76 \text{ units}$$

$$EOQ_{60} = \sqrt{\frac{2(936)(45)}{0.25(60)}} = 75 \text{ units}$$

Calculating total cost at each price break price

$$C_{75} = \frac{75}{2}[(0.25)(₹60)] + \frac{936}{75}(₹45) + (₹60)(936)$$

$$\boldsymbol{C_{75} = ₹57284}$$

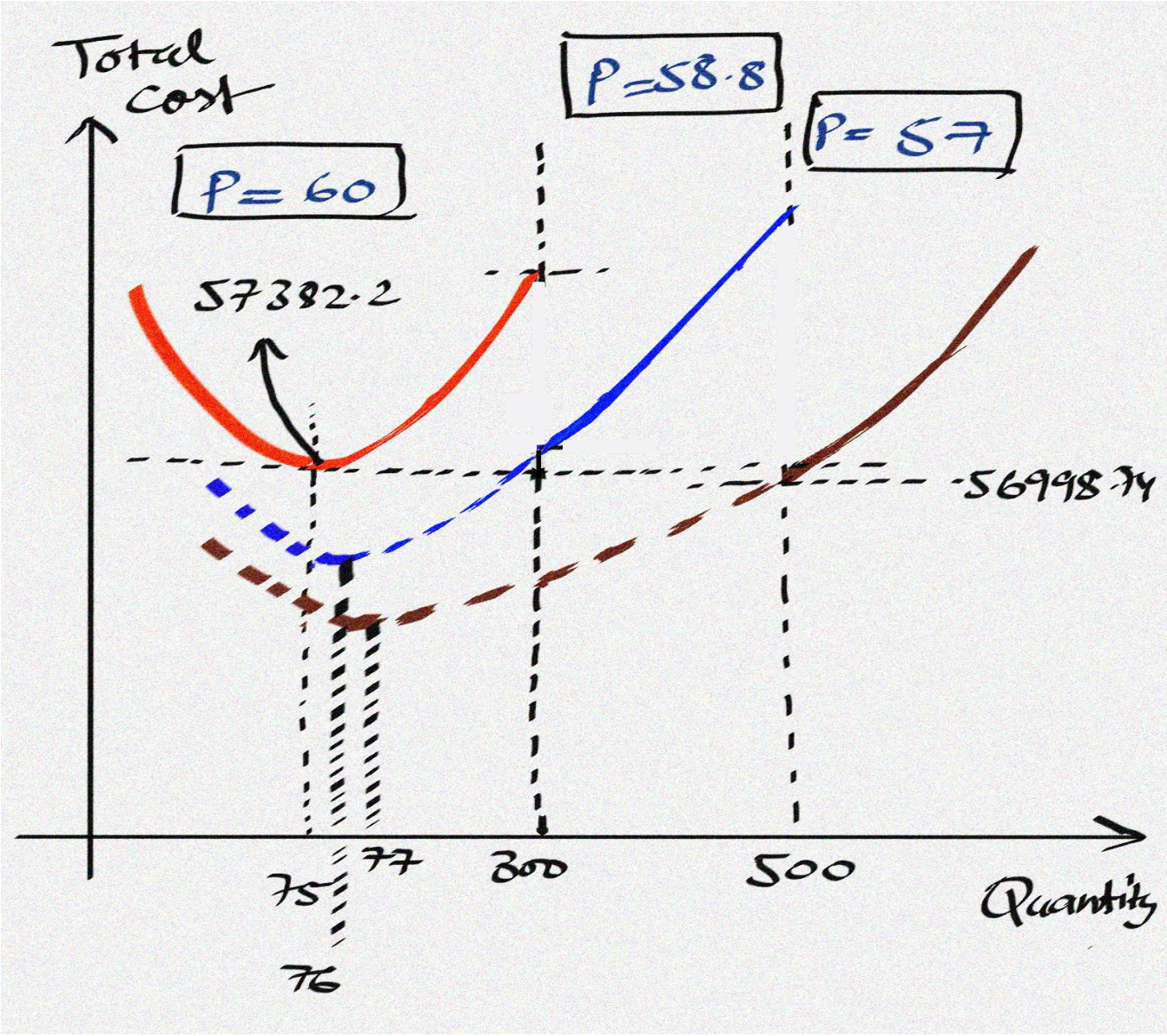
$$C_{300} = \frac{300}{2}[(0.25)(₹58.8)] + \frac{936}{300}(₹45) + (₹58.8)(936)$$

$$\boldsymbol{C_{300} = ₹57382.2}$$

$$C_{500} = \frac{500}{2}[(0.25)(₹57)] + \frac{936}{500}(₹45) + (₹57)(936)$$

$$\boldsymbol{C_{500} = ₹56998.74}$$

- Minimum Total Cost occurs at Q=500

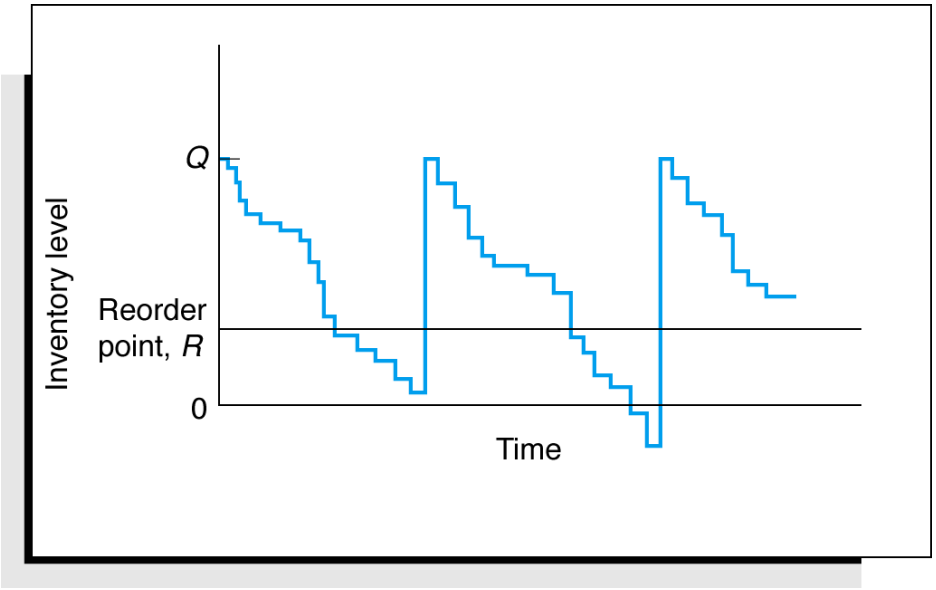


Probabilistic Models

"A statistical model applicable when Product demand (or) lead time (or) any other variable is not known but can be specified by means of a probability distribution."

Continuous System

- Inventory level might be depleted at slower or faster rate during lead time.
- When demand is uncertain, safety stock is added as a hedge against stock-out.
- Focus must be on demand distribution during the lead time

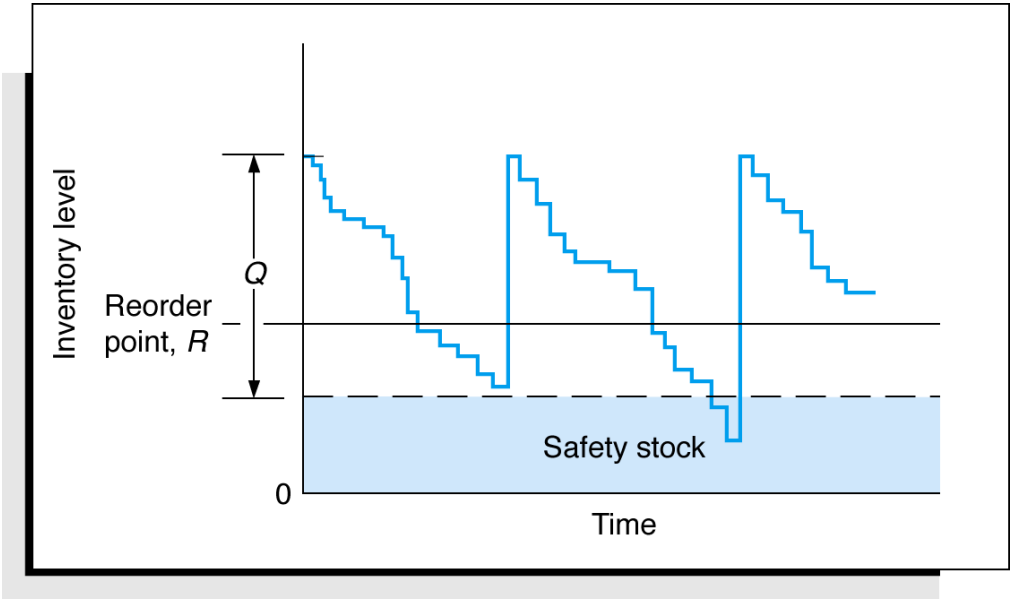


Reorder Point Quantity

- Under deterministic conditions, when both demand and lead time are constant, the reorder point is set equal to lead time demand.

- Under probabilistic conditions, when demand and/or lead time varies, the reorder point often includes safety stock

- <u>Safety stock</u> is the amount by which the reorder point exceeds the expected (average) lead time demand.



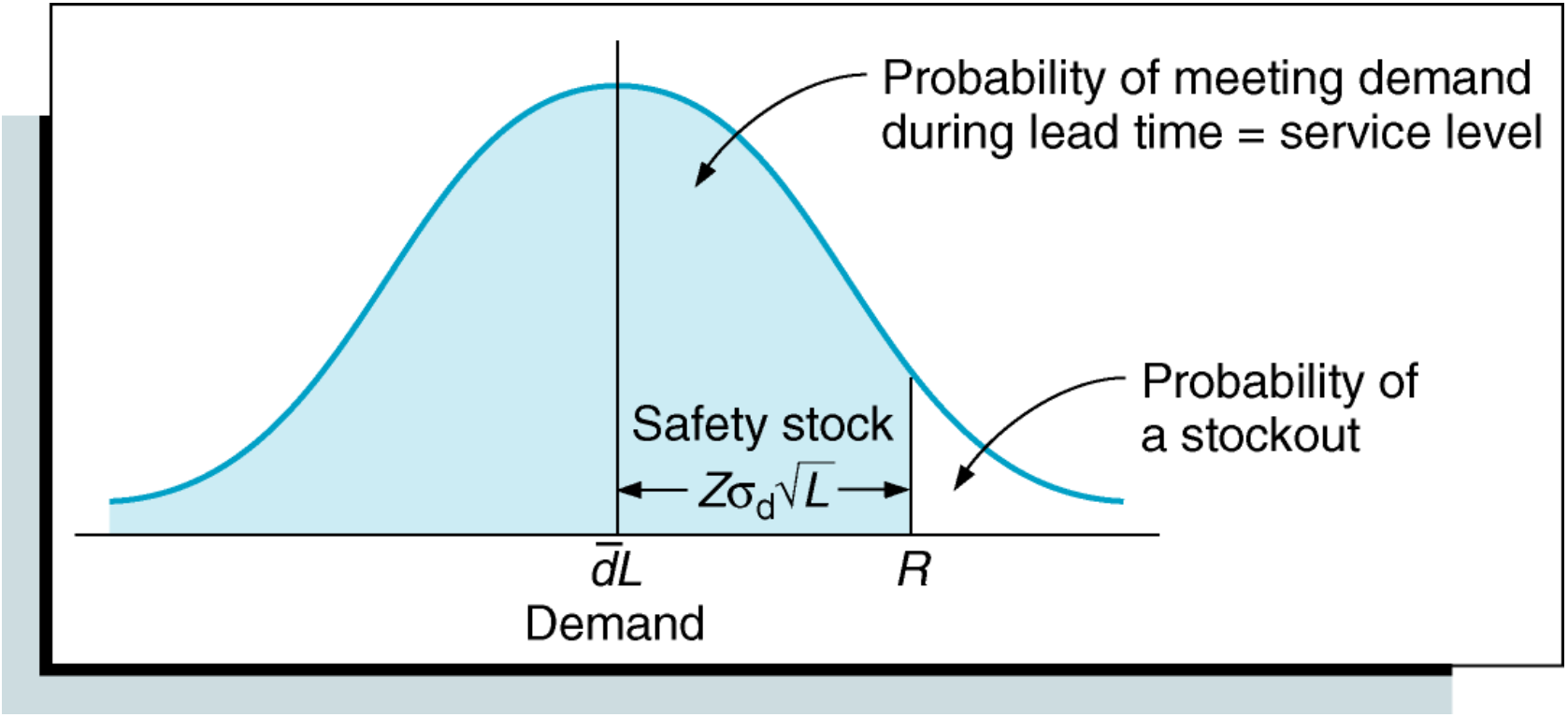
Safety Stock and Service Level

- Safety stock reduces the chances of a stockout during lead time
- The complement of this chance is called the service level
- Service level is defined as the probability of not incurring a stockout during any one lead time
- The higher the probability inventory will be on hand, the more likely customer demand will be met.
- Service level of 90% means there is a .90 probability that demand will be met during lead time and .10 probability of a stockout.

Reorder Point

- Assumptions
  - Lead-time demand is normally distributed with mean μ and standard deviation σ.
  - Approximate optimal order quantity: EOQ
  - Service level is defined in terms of the probability of no stockouts during lead time and is reflected in z.
  - Shortages are not backordered.
  - Inventory position is reviewed continuously.

Reorder Point with Variable Demand



Reorder Point with Variable Demand

$$R = \bar{d}L + Z\sigma_d\sqrt{L}$$

where:

R = reorder point

$\bar{d}$ = average daily demand

L = lead time

σ_d = the standard deviation of daily demand

Z = number of standard deviations corresponding to service level probability

$Z\sigma_d\sqrt{L}$ = safety stock

Reorder Point with Variable Lead Time

- For constant demand and variable lead time:

$$R = d\bar{L} + Zd\sigma_L$$

where:

d = constant daily demand
$\bar{L}$ = average lead time
σ_L = standard deviation of lead time
$d\sigma_L$ = standard deviation of demand during lead time
$Zd\sigma_L$ = safety stock

Reorder Point
Variable Demand and Lead Time

- When both demand and lead time are variable:

$$R = \bar{d}\bar{L} + Z\sqrt{(\sigma_d)^2\bar{L} + (\sigma_L)^2\bar{d}^2}$$

where:

$\bar{d}$ = average daily demand

$\bar{L}$ = average lead time

$\sqrt{(\sigma_d)^2\bar{L} + (\sigma_L)^2\bar{d}^2}$ = standard deviation of demand during lead time

$Z\sqrt{(\sigma_d)^2\bar{L} + (\sigma_L)^2\bar{d}^2}$ = safety stock

Periodic Review System (P)

- In a P system, the inventory position (IP) is not reviewed continuously, but once every T units of time. Then, an order may be placed to replenish the inventory.

- The ordering quantity depends on the inventory policy followed.

- If (R, Q) (continuous review system) policy is followed, the ordering quantity is equal to the calculated EOQ value (Q).

- If (s, S) policy (periodic review system) is followed, the ordering quantity is equal to the difference between the target inventory level (order upto level, S) and the inventory position (IP) at the time of review.

Periodic Review System (P)

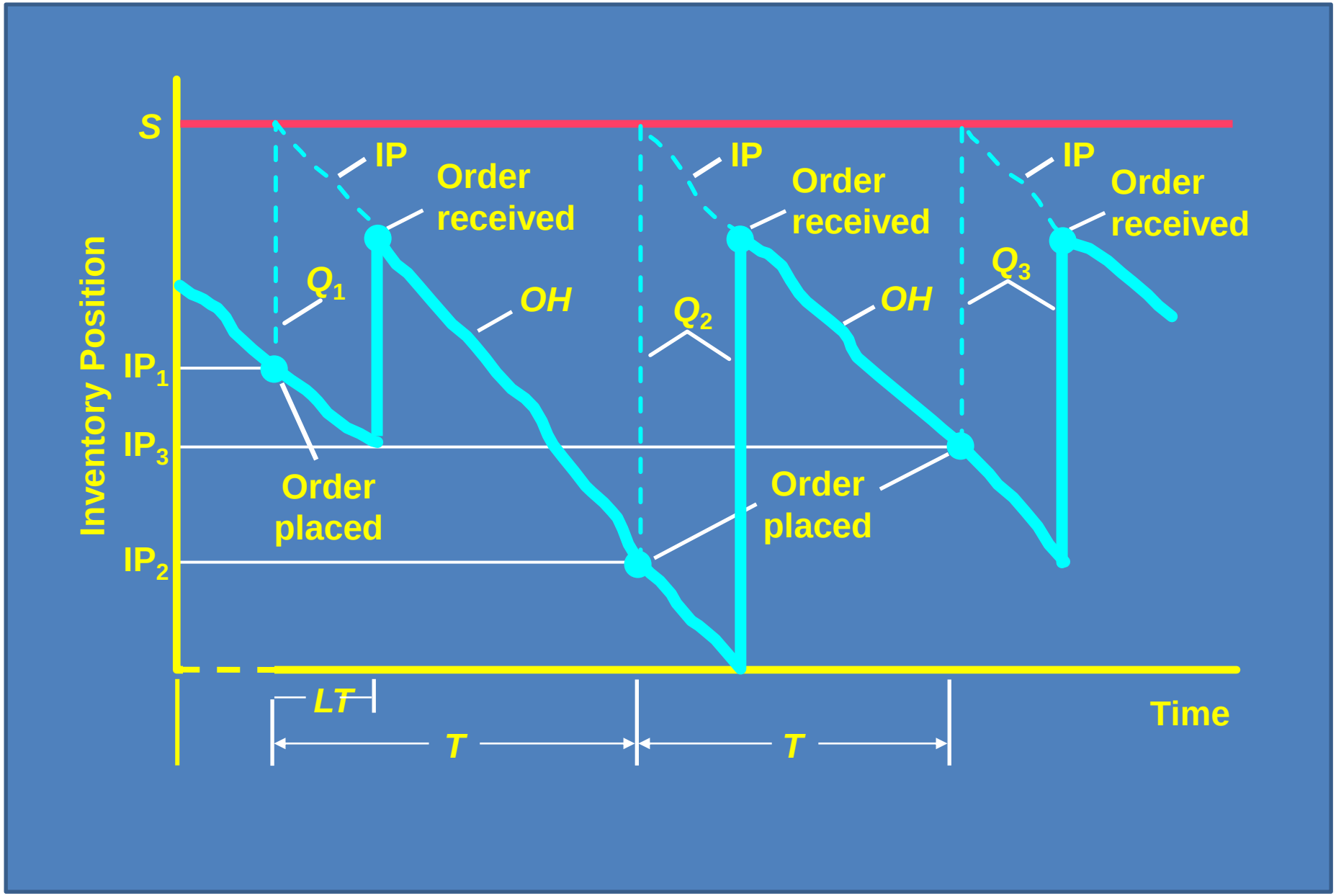
- In the (s, S) policy, an order is placed only when the inventory position is at or below the reorder point (s) at the time of review.

- In the case of a base stock policy (S policy), there is no defined reorder point and hence, an order is placed after <u>every</u> review to bring the inventory position upto S.

- As $T \to 0$, a P system becomes quite similar to a Q system.

- Here, demand and lead time may be deterministic or stochastic.

Periodic Review System (P) for an S policy



How Much to Order in a P System

EXAMPLE

A distribution center has a backorder (BO) for five 55-inch LED TVs. No inventory is currently on hand (OH), and now is the time to review. How many should be reordered if S = 400 and no receipts are scheduled (SR)?

SOLUTION

$$IP = OH + SR - BO$$

$$= 0 + 0 - 5 = -5 \text{ sets}$$

$$S - IP = 400 - (-5) = 405 \text{ sets}$$

That is, 405 sets must be ordered to bring the inventory position up to S sets.

Periodic Review System

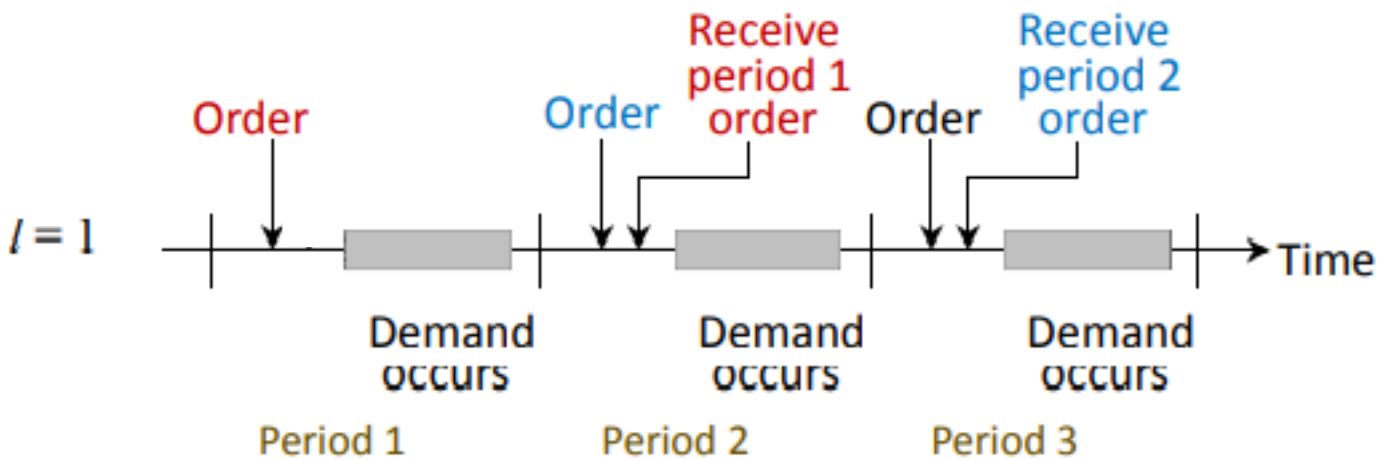
- Selecting the period of time between reviews (T)
- The order-up-to level (S) when demand is variable and lead time is constant will be equal to the average demand during the protection period ($T + L$) + Safety Stock

$$S = \bar{d}(T + L) + \text{safety stock for protection interval}$$

$$\text{Safety stock(SS)} = Z \times \sigma_d \sqrt{T + L}$$

- The formula for order-up-to level and safety stock may have to be adjusted depending upon the <u>sequence</u> in which inventory events occur.

An Example for Event Sequence



Event Sequences in Inventory

- Consider a periodic review inventory system with period length T, mean demand $\bar{d}$ and a constant lead time L.

- Suppose that an order is placed at the end of period T. Now this order will be fulfilled in period $T + L$.

- This replenishment can occur either before the demand in period $T + L$ is realized or after it is realized.

Event Sequences in Inventory

- Depending on the sequence in which inventory events occur, the effective lead time may actually be one more or one less than the actual lead time.

- In general, the lead time interval in a P-system should correspond to the number of times the demand is realized between order placement and receipt of items.

- This is illustrated with an example.

Case 1: Lead Time = No. of demand realizations between order placement and order receipt

In this case, there is no need for any adjustment in the order-up-to level formula.

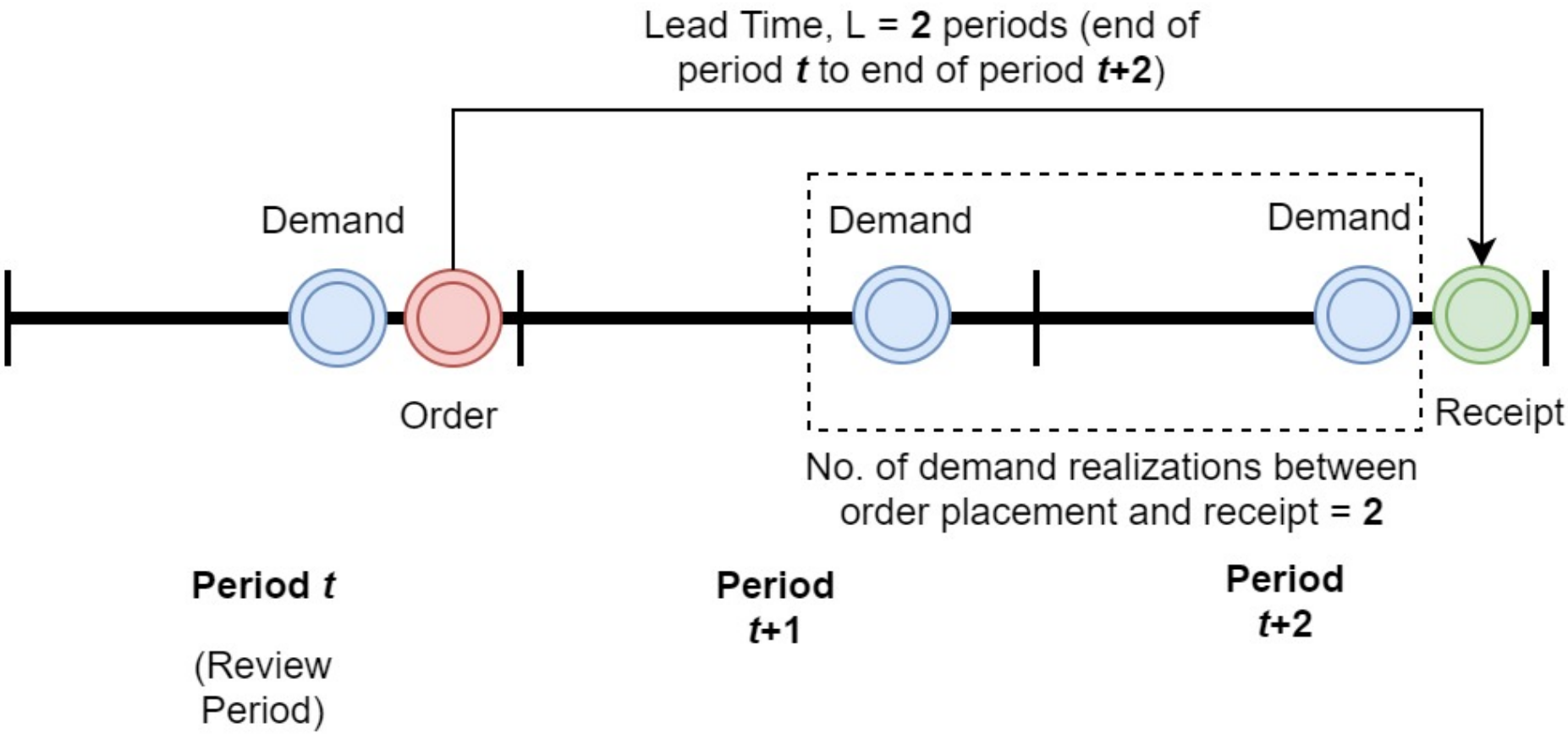
Mean Demand (μ)	10	
Review Period (T)	3	
Lead Time (L)	2	
Order Up To Level (S)	50	$S = \mu(T + L)$
Sequence of Events:		

Demand realization occurs at the end of a period

Inventory is reviewed once every T periods (at the end of the respective period). This review occurs **after** the realization of demand in that period.

Replenishment occurs at the end of periods **after** the realization of demand.

Sequence of events in Case 1



State	Period (t)	Event	Demand	Inv. Pos	Inv. On Hand	Receipts	Order Quantity
Initial	0			50	50		
Beginning of Period	1			50	50		
End of Period		Demand Realization	10	40	40		
Beginning of Period	2			40	40		
End of Period		Demand Realization	10	30	30		
Beginning of Period	3			30	30		
End of Period		Demand Realization	10	20	20		
		Review		50	20		30
Beginning of Period	4			50	20		
End of Period		Demand Realization	10	40	10		
Beginning of Period	5			40	10		
End of Period		Demand Realization	10	30	0		
		Order Receipt		30	30	30	
Beginning of Period	6			30	30		
End of Period		Demand Realization	10	20	20		
		Review		50	20		30
Beginning of Period	7			50	20		
End of Period		Demand Realization	10	40	10		
Beginning of Period	8			40	10		
End of Period		Demand Realization	10	30	0		
		Order Receipt		30	30	30	
Beginning of Period	9			30	30		
End of Period		Demand Realization	10	20	20		
		Review		50	20		30
Beginning of Period	10			50	20		
End of Period		Demand Realization	10	40	10		
Beginning of Period	11			40	10		
End of Period		Demand Realization	10	30	0		
		Order Receipt		30	30	30	
Beginning of Period	12			30	30		
End of Period		Demand Realization	10	20	20		
		Review		50	20		30

Calculations for Case 1

Case 2: Lead Time > No. of demand realizations between order placement and order receipt

In this case, L is replaced by $L - 1$ in the order-up-to level formula.

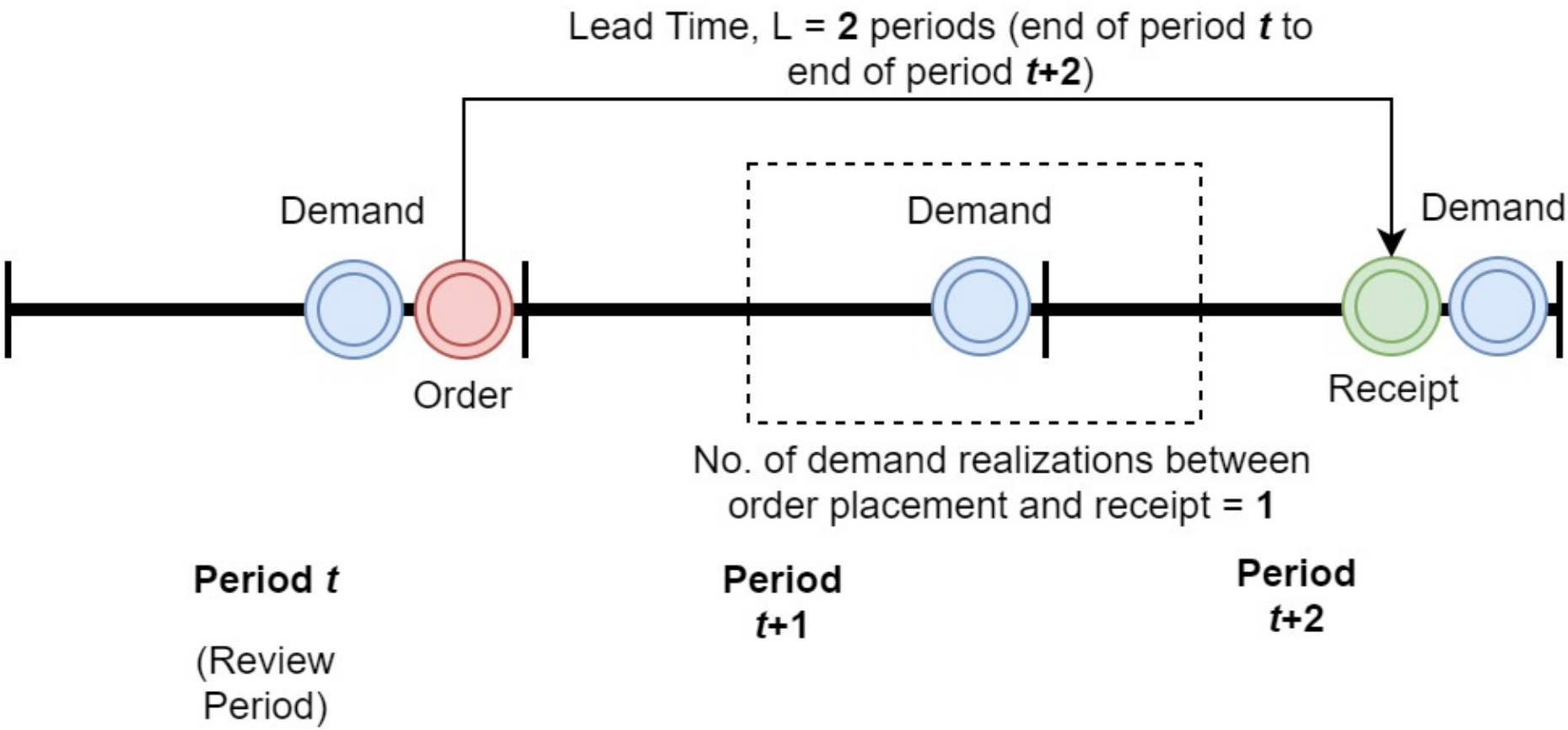
Mean Demand (μ)	10	
Review Period (T)	3	
Lead Time (L)	2	
Order Up To Level (S)	40	$S = \mu(T + L - 1)$
Sequence of Events:		

Demand realization occurs at the end of a period

Inventory is reviewed once every T periods(at the end of the respective period). This review occurs **after** the realization of demand in that period.

Replenishment occurs at the end of periods **before** the realization of demand.

Sequence of events in Case 2



State	Period (t)	Event	Demand	Inv. Pos	Inv. On Hand	Receipts	Order Quantity
Initial	0			40	40		
Beginning of Period	1			40	40		
End of Period		Demand Realization	10	30	30		
Beginning of Period	2			30	30		
End of Period		Demand Realization	10	20	20		
Beginning of Period	3			20	20		
End of Period		Demand Realization	10	10	10		
		Review		40	10		30
Beginning of Period	4			40	10		
End of Period		Demand Realization	10	30	0		
Beginning of Period	5			30	0		
End of Period		Order Receipt		30	30	30	
		Demand Realization	10	20	20		
Beginning of Period	6			20	20		
End of Period		Demand Realization	10	10	10		
		Review		40	10		30
Beginning of Period	7			40	10		
End of Period		Demand Realization	10	30	0		
Beginning of Period	8			30	0		
End of Period		Order Receipt		30	30	30	
		Demand Realization	10	20	20		
Beginning of Period	9			20	20		
End of Period		Demand Realization	10	10	10		
		Review		40	10		30
Beginning of Period	10			40	10		
End of Period		Demand Realization	10	30	0		
Beginning of Period	11			30	0		
End of Period		Order Receipt		30	30	30	
		Demand Realization	10	20	20		
Beginning of Period	12			20	20		
End of Period		Demand Realization	10	10	10		
		Review		40	10		30

Calculations for Case 2

Case 3: Lead Time < No. of demand realizations between order placement and order receipt

In this case, L is replaced by $L + 1$ in the order-up-to level formula.

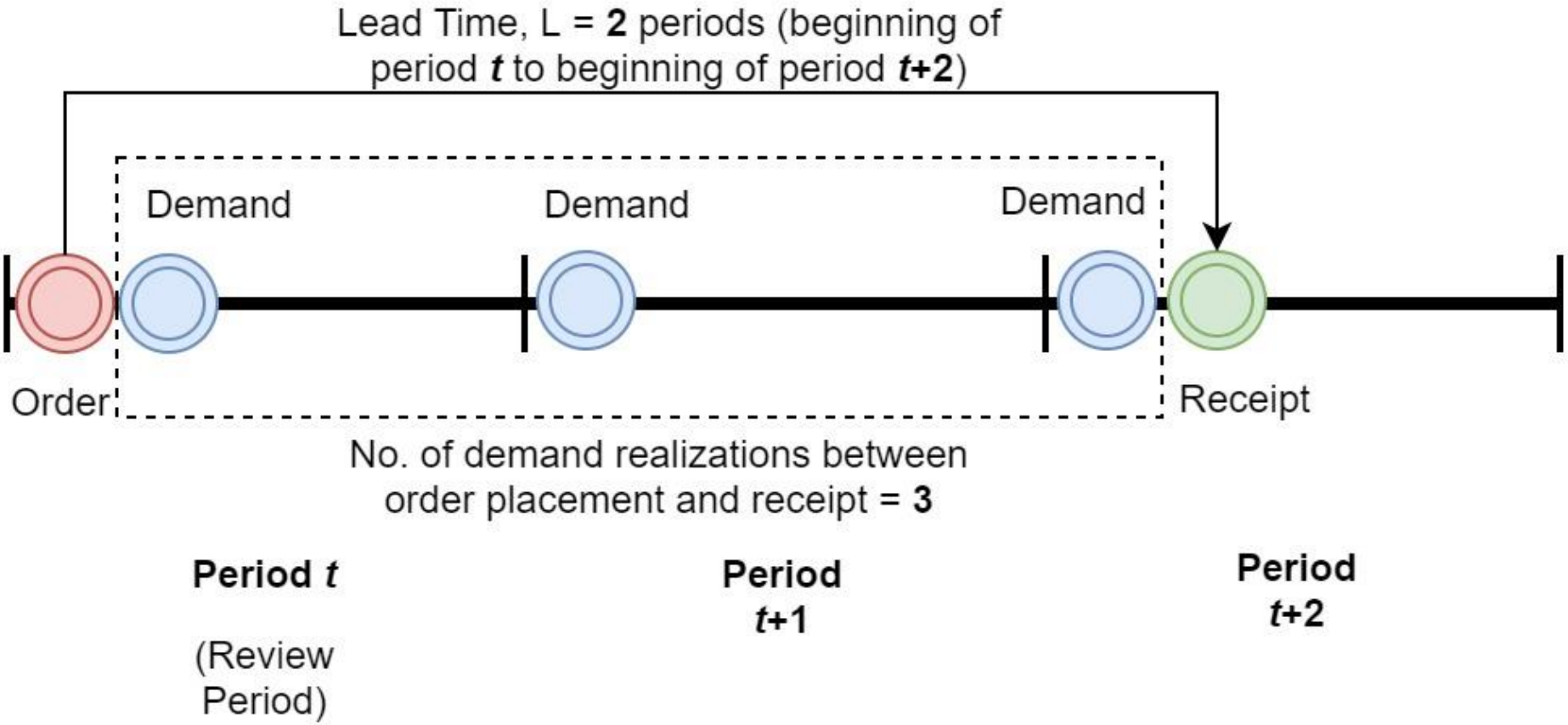
Mean Demand (μ)	10	
Review Period (T)	3	
Lead Time (L)	2	
Order Up To Level (S)	60	$S = \mu(T + L + 1)$
Sequence of Events:		

Demand realization occurs at the beginning of a period

Inventory is reviewed once every T periods (at the beginning of the respective period). This review occurs **before** the realization of demand in that period.

Replenishment occurs at the beginning of periods **after** the realization of demand.

Sequence of events in Case 3



State	Period (t)	Event	Demand	Inv. Pos	Inv. On Hand	Receipts	Order Quantity
Initial	0			60	60		
Beginning of Period	1	Demand Realization	10	50	50		
End of Period				50	50		
Beginning of Period	2	Demand Realization	10	40	40		
End of Period				40	40		
Beginning of Period	3	Demand Realization	10	30	30		
End of Period				30	30		
Beginning of Period	4	**Review**		60	30		30
		Demand Realization	10	50	20		
End of Period				50	20		
Beginning of Period	5	Demand Realization	10	40	10		
End of Period				40	10		
Beginning of Period	6	Demand Realization	10	30	0		
		Order Receipt		30	30	30	
End of Period				30	30		
Beginning of Period	7	**Review**		60	30		30
		Demand Realization	10	50	20		
End of Period				50	20		
Beginning of Period	8	Demand Realization	10	40	10		
End of Period				40	10		
Beginning of Period	9	Demand Realization	10	30	0		
		Order Receipt		30	30	30	
End of Period				30	30		
Beginning of Period	10	**Review**		60	30		30
		Demand Realization	10	50	20		
End of Period				50	20		
Beginning of Period	11	Demand Realization	10	40	10		
End of Period				40	10		
Beginning of Period	12	Demand Realization	10	30	0		
		Order Receipt		30	30	30	
End of Period				30	30		

Calculations for Case 3

Periodic Review

Advantages

- Orders placed at fixed intervals
- No continuous inventory count
- Convenient administratively
- Useful when vendors visit routinely
- If items are ordered from same supplier may yield savings on ordering and shipping costs

Disadvantages

- Possibility of stock out between intervals
- High level of safety stock is needs to be maintained.
- Increased Carrying costs
- Cost of periodic reviews

Comparison of Q and P Systems

P Systems

- **Convenient to administer**
- **Orders may be combined**
- **Suitable for items with high demand**

Q Systems

- **Individual review frequencies**
- **Possible quantity discounts**
- **Lower, less - expensive safety stocks**
- **Suitable for items with low demand**

Optional

Dynamic Lot Sizing

- Consider a multi-period inventory model in which the demand is deterministic but time-varying.

- When dealing with lot sizing for time-varying demand, it is generally assumed that there is a finite number of discrete time steps, or periods.

- The demand in each period is known, and for simplicity, it is assumed that the period demand takes place at the beginning of the period.

Optional

Dynamic Lot Sizing

- Since the demand is varying over time, the optimal lot sizes need not be constant over the time periods.

- The objective is to determine the points of time at which orders should be placed and the quantities to be ordered such that the total cost across all periods is minimized.

- This dynamic lot sizing problem may be solved by exact methods such as Wagner-Whitin Algorithm or by approximate methods such as Silver-Meal Heuristic.

Optional

Wagner-Whitin Algorithm

- The Wagner-Whitin Algorithm was developed by Wagner and Whitin in the year 1958.

- It uses a **dynamic programming** approach to recursively determine the optimal lot sizes.

- The most basic form of the W-W algorithm assumes **zero lead time**, **infinite fill rate** and **no backlogging**, i.e., the inventory is replenished by an amount equal to the ordered quantity immediately after an order is placed, and unmet demands cannot be backlogged for fulfillment in later periods.

Optional

Wagner-Whitin Algorithm

According to Wagner and Whitin, an optimal solution to a classic dynamic lot size problem must satisfy the following two conditions:

1. A delivery should take place only when the inventory is zero.

2. The holding cost for a period demand should never exceed the ordering cost.

Let,

T - No. of periods
d_t - Demand in period t
A_t - Ordering cost (Setup cost) per order in period t
h_t - Holding cost per unit in period t

Optional

Wagner-Whitin Algorithm

Z_k = minimum cost incurred in satisfying demands over periods 1, 2,..., k, i.e., when we disregard periods k + 1, k + 2, ..., T;

$C_{t,k}$ = cost incurred in satisfying demands over periods t, t + 1, t + 2, ..., k, given that the inventory is to be replenished at the *beginning* of period t ($1 \leq t \leq k$), just before the demand in that period is realized. (The period demand is also assumed to occur at the beginning of a period.)

$C_{t,k}$ can be calculated as follows:

$$C_{t,k} = A_t + \sum_{i=t}^{k-1} h_i \sum_{j=i+1}^{k} d_j$$

Optional

Wagner-Whitin Algorithm

- From the optimality conditions specified by Wagner and Whitin, the holding cost for a period demand must not exceed the ordering cost in that period.

- Using above condition, $C_{t,k}$ is redefined as:

$$C_{t,k} = \begin{cases} A_t + \sum_{i=t}^{k-1} h_i \sum_{j=i+1}^{k} d_j \,, & if\ A_k \geq \sum_{i=t}^{k-1} h_i d_k \\ \infty \,, & otherwise \end{cases}$$

- Now, $C_{t,k}$ covers the costs of satisfying demands from periods t to k assuming procurement at the beginning of period t, while Z_{t-1} covers the minimum cost of satisfying demands from the first period upto period $t-1$.

Optional

Wagner-Whitin Algorithm

- Using these two terms, Z_k can be calculated recursively as follows:

$$Z_k = \min_{1 \le t \le k} \{Z_{t-1} + C_{t,k}\}$$

- Obviously, $Z_0 = 0$ and $Z_1 = C_{1,1} = A_1$, since with just one period, we get an ordering cost but no holding costs.

- Z_2 can be calculated once Z_1, $C_{1,2}$ and $C_{2,2}$ are known.

$$Z_2 = \min\{Z_0 + C_{1,2}\ ,\ Z_1 + C_{2,2}\}$$

- By extension, Z_k can be determined for all $k \le T$.

Optional

Wagner-Whitin Algorithm

- To find the optimal lot sizes, the following procedure is carried out.

- First, considering $k = T$, calculate Z_k. The minimum value is the optimal cost, and the minimizing t is the period with the last delivery.

- The cumulative sum of demands from periods t to T represent the quantity to be ordered in period t. Denote the minimizing t by t'.

Optional

Wagner-Whitin Algorithm

- Now, again consider $k = t' - 1$ and calculate Z_k. Find the period t'' at which the total cost upto period $t' - 1$ is minimum and add up the demands from periods t'' to $t' - 1$ to determine the ordering quantity.

- Next, consider $k = t'' - 1$, and repeat the procedure until the minimizing t equals one, since all periods with deliveries would have been obtained then.

- Since the ordering quantity in any period is exactly equal to the sum of demands from that period until the period before the one in which the next order is to be placed, we see that the procedure implicitly satisfies the first condition for optimality.

Optional

Example

The demands in $T = 10$ periods are given in the following table.

The ordering cost is $A = ₹300$ for all periods and the holding cost is $h = ₹1$ per unit in each period. Find the optimal lot sizes using Wagner-Whitin algorithm.

Period t	1	2	3	4	5	6	7	8	9	10
d_t	50	60	90	70	30	100	60	40	80	20

Optional

Calculation

$Z_0 = ₹\mathbf{0}; Z_1 = C_{1,1} = A_1 = ₹\mathbf{300};$

$$Z_2 = \min\{Z_0 + C_{1,2}\ , Z_1 + C_{2,2}\}$$

Now, $Z_0 + C_{1,2} = Z_0 + A_1 + h_1 d_2$

$$Z_0 + C_{1,2} = 0 + 300 + (1)(60) = ₹\mathbf{360}$$

Similarly, $Z_1 + C_{2,2} = Z_1 + A_2$

$$Z_1 + C_{2,2} = 300 + 300 = ₹\mathbf{600}$$

Therefore, $Z_2 = \min\{360, 600\} = ₹\mathbf{360}$

Optional

Calculation

$$Z_3 = \min\{Z_0 + C_{1,3}\ ,\ Z_1 + C_{2,3}\ ,\ Z_2 + C_{3,3}\}$$

Now, $Z_0 + C_{1,3} = Z_0 + A_1 + h_1(d_2 + d_3) + h_2 d_3$

$$Z_0 + C_{1,3} = 0 + 300 + (1)(60 + 90) + (1)(90) = \textbf{₹540}$$

Similarly, $Z_1 + C_{2,3} = Z_1 + A_2 + h_2 d_3$

$$Z_1 + C_{2,2} = 300 + 300 + (1)(90) = \textbf{₹690}$$

Similarly, $Z_2 + C_{3,3} = Z_2 + A_3$

$$Z_1 + C_{2,2} = 360 + 300 = \textbf{₹660}$$

Therefore, $Z_3 = \min\{540, 690, 660\} = \textbf{₹540}$

Optional

Calculation

Similarly, Z_4 and Z_5 are calculated as ₹$\mathbf{730}$ and ₹$\mathbf{790}$, respectively.

While calculating $Z_0 + C_{1,6}$ in order to find the minimum value of Z_6, we encounter the first such instance in this problem where the holding cost for a period demand exceeds the ordering cost, i.e.,

$$\sum_{i=1}^{5} h_i d_6 = (1+1+1+1+1)(100) = 500 > A_6 = 300$$

Therefore, from the definition of $C_{t,k}$, $C_{1,6}$ and consequently $Z_0 + C_{1,6}$ are set to infinity. By extension, all $C_{1,k}$ for k in the range $6 \leq k \leq T$ can be set to infinity, and hence, they need not be considered when computing Z_k for the same range of values of k.

Optional

Calculation

By continuing in this vein, the $Z_{t-1} + C_{t,k}$ values for all possible values of t and k have been calculated and tabulated as follows:

Period t	1	2	3	4	5	6	7	8	9	10
d_t	50	60	90	70	30	100	60	40	80	20
$k=t$	300	600	660	840	1030	1090	1370	1450	1530	1770
$k=t+1$	360	690	730	870	1130	1150	1410	1530	1550	–
$k=t+2$	540	830	790	1070	1250	1230	1570	1570	–	–
$k=t+3$	750	920	1090	1250	1370	1470	1630	–	–	–
$k=t+4$	870	–	1330	1410	–	1550	–	–	–	–
$k=t+5$	–	–	1530	–	–	–	–	–	–	–

The value of k is constant across a diagonal (as marked in the above table). The minimum of the entries along a diagonal will give the corresponding Z_k value.

Optional

Calculation

- The optimal cost is the value of Z_k at $k = T = 10$. Considering the last diagonal from the previous table,

$$Z_{10} = \min\{1550, 1630, 1570, 1550, 1770\} = ₹\mathbf{1550}$$

- To determine the optimal lot sizes, find the period t at which $Z_{t-1} + C_{t,10}$ is minimum, i.e., equal to Z_{10}.

- Here, the same minimum value is obtained for two different values of t (9 and 6). So, two alternative optimal solutions are possible.

Optional

Calculation

- First, consider the solution with $t = 9$. Here, the inventory is to be replenished at the beginning of period 9 to satisfy the demands in periods 9 and 10.

- Therefore, the total ordering quantity in period 9 will be the sum of demands in periods 9 and 10.

$$Q_9 = d_9 + d_{10} = 80 + 20 = \mathbf{100\ units}$$

- Next, set $k = 9 - 1 = 8$, and find the minimum cost upto period 8 (Z_8).

$$Z_8 = \min\{1530, 1410, 1370, 1230, 1410, 1450\} = ₹\mathbf{1230}$$

Optional

Calculation

- Then, find the period t at which $Z_{t-1} + C_{t,8}$ is minimum, i.e., equal to Z_8.

- Here, the minimum cost occurs at $t = 6$ and therefore, the ordering quantity will be the sum of demands from periods 6 to 8.

$$Q_6 = d_6 + d_7 + d_8 = 100 + 60 + 40 = \mathbf{200\ units}$$

- Next, set $k = 6 - 1 = 5$, and find the minimum cost upto period 5 (Z_5).

$$Z_5 = \min\{870, 920, 790, 870, 1030\} = \text{₹}\mathbf{790}$$

Optional

Calculation

- Then, find the period t at which $Z_{t-1} + C_{t,5}$ is minimum, i.e., equal to Z_5.

- Here, the minimum cost occurs at $t = 3$ and therefore, the ordering quantity will be the sum of demands from periods 3 to 5.

$$Q_3 = d_3 + d_4 + d_5 = 90 + 70 + 30 = \mathbf{190\ units}$$

- Next, set $k = 3 - 1 = 2$, and find the minimum cost upto period 2 (Z_2).

$$Z_2 = \min\{360, 660\} = ₹\mathbf{360}$$

Optional

Calculation

- Then, find the period t at which $Z_{t-1} + C_{t,2}$ is minimum, i.e., equal to Z_2.

- Here, the minimum cost occurs at $t = 1$ and therefore, the ordering quantity will be the sum of demands in periods 1 and 2.

$$Q_1 = d_1 + d_2 = 50 + 60 = \mathbf{110\ units}$$

- Now, we have reached $t = 1$. This implies that the demands in all periods have been considered and the optimal lot sizes have been determined. Hence, the process terminates.

Optional

Example

- In a similar way, the optimal lot sizes corresponding to the alternative optimal solution can be determined.

- The optimal lot sizes for both solutions are summarized in the table below:

Period t	1	2	3	4	5	6	7	8	9	10
Solution 1	110	–	190	–	–	200	–	–	100	–
Solution 2	110	–	190	–	–	300	–	–	–	–

Optional

Wagner-Whitin Algorithm with Backlogging

- Zangwill in the year 1969 extended the Wagner-Whitin algorithm to incorporate backlogging, which allows unfulfilled demands to be fulfilled at later periods with an additional penalty.

- The procedure is similar to the basic W-W algorithm, with the only major difference being the manner in which the cost term $C_{t,k}$ is defined.

- Let s_t be the unit cost of shortage in period t, while other notations largely remain the same as in the basic W-W algorithm.

Optional

Wagner-Whitin Algorithm with Backlogging

- In the basic W-W algorithm, $C_{t,k}$ represented the cost associated with satisfying the demands in periods $t, t+1, t+2, \ldots, k$ given that the inventory is replenished at the beginning of period t.

- Since no backlogging is permitted in the basic W-W algorithm, the inventory is replenished at the beginning of period t if the inventory position at the end of period $t-1$ becomes zero.

Optional

Wagner-Whitin Algorithm with Backlogging

- However, when backlogging is permitted, the inventory position can be allowed to become negative as long as it facilitates a reduction in the total cost.

- This means, when we consider the cost of satisfying demands over multiple periods, say from period t to period k, there may be some period u $(t \leq u \leq k)$ between t and k in which replenishing the inventory leads to minimum cost.

- Now, we define this cost term $(C_{t,u,k})$ using three indices as follows:

Optional

Wagner-Whitin Algorithm with Backlogging

$C_{t,u,k}$ = cost of satisfying the demands in periods $t, t+1, t+2, \ldots, k$ given that the inventory is replenished at the beginning of period u ($t \leq u \leq k$).

$$C_{t,u,k} = A_u + \sum_{i=u}^{k-1} h_i \sum_{j=i+1}^{k} d_j + \sum_{i=t}^{u-1} s_i \sum_{j=t}^{i} d_j$$

In other words, if the inventory is replenished at some period u between t and k, the total cost will include the ordering cost, the shortage costs incurred for the unfulfilled demands from period t to period $u-1$, and the holding costs incurred in stocking items to satisfy demands from period $u+1$ to period k.

Wagner-Whitin Algorithm with Backlogging

- Let $C_{t,k}$ = minimum cost of satisfying the demands in periods $t, t+1, t+2, \ldots, k$ given that the inventory is replenished at some period u between t and k ($t \leq u \leq k$).

$$C_{t,k} = \min_{t \leq u \leq k} \{C_{t,u,k}\}$$

- The minimum cost of satisfying demands from period 1 upto period k can be recursively defined as:

$$Z_k = \min_{1 \leq t \leq k} \{Z_{t-1} + C_{t,k}\}$$

Optional

Wagner-Whitin Algorithm with Backlogging

- The value of Z_k at $k = T$ gives the minimum total cost at which the demands across the planning period can be fulfilled.

- The process of determining the ordering periods and lot sizes is similar to that in the basic W-W algorithm, but with one additional step.

- Here, once the period t at which the minimum value of Z_k occurs is determined, the next step is to determine the period u between t and k at which the minimum value of $C_{t,k}$ occurs.

Optional

Wagner-Whitin Algorithm with Backlogging

- Thus, the period u will be the ordering period, and the sum of demands from period t to period k will be the ordering quantity.

- In other words, identifying the value of t allows grouping of periods for a single replenishment decision, while identifying u helps to decide the period within that group of periods in which the replenishment is to take place, such that the total cost is minimized.

Optional

Optional

Example

Period	Demand (units)	Ordering Cost (₹)	Holding Cost per unit (₹)	Shortage Cost per unit (₹)
1	10	50	2	3
2	40	20	2	3
3	20	30	1	3
4	10	50	1	3

Find the optimal lot sizes assuming backlogging is allowed.

Optional

Calculation

$Z_0 = ₹\mathbf{0}; Z_1 = Z_0 + C_{1,1} = C_{1,1,1} = A_1 = ₹\mathbf{50};$

$$Z_2 = \min\{Z_0 + C_{1,2}\ , Z_1 + C_{2,2}\}$$

Now, $C_{1,2} = \min\{C_{1,1,2}\ , C_{1,2,2}\}$

$$C_{1,1,2} = A_1 + h_1 d_2 = 50 + (2)(40) = ₹\mathbf{130}$$

$$C_{1,2,2} = A_2 + s_1 d_1 = 20 + (3)(10) = ₹\mathbf{50}$$

Therefore, $C_{1,2} = \min\{130, 50\} = ₹\mathbf{50}$

Now, $C_{2,2} = C_{2,2,2} = A_2 = ₹\mathbf{20}$

$$\therefore\ Z_2 = \min\{0 + 50, 50 + 20\} = \min\{50, 70\} = ₹\mathbf{50}$$

Optional

Calculation

$$Z_3 = \min\{Z_0 + C_{1,3}\ , Z_1 + C_{2,3}\ , Z_2 + C_{3,3}\}$$

Now, $C_{1,3} = \min\{C_{1,1,3}\ , C_{1,2,3}\ , C_{1,3,3}\}$

$$C_{1,1,3} = A_1 + h_1(d_2 + d_3) + h_2 d_3 = 50 + (2)(40 + 20) + (2)(20) = \text{₹}\mathbf{210}$$

$$C_{1,2,3} = A_2 + h_2 d_3 + s_1 d_1 = 20 + (2)(20) + (3)(10) = \text{₹}\mathbf{90}$$

$$C_{1,3,3} = A_3 + s_1 d_1 + s_2(d_1 + d_2) = 30 + (3)(10) + (3)(10 + 40) = \text{₹}\mathbf{210}$$

Therefore, $C_{1,3} = \min\{210, 90, 210\} = \text{₹}\mathbf{90}$

Now, $C_{2,3} = \min\{C_{2,2,3}\ , C_{2,3,3}\}$

Optional

Calculation

$$C_{2,2,3} = A_2 + h_2 d_3 = 20 + (2)(20) = ₹\mathbf{60}$$

$$C_{2,3,3} = A_3 + s_2 d_2 = 30 + (3)(40) = ₹\mathbf{150}$$

Therefore, $C_{2,3} = \min\{60, 150\} = ₹\mathbf{60}$

Now, $C_{3,3} = C_{3,3,3} = A_3 = ₹\mathbf{30}$

$\therefore\ Z_3 = \min\{0 + 90, 50 + 60, 50 + 30\} = \min\{90, 110, 80\} = ₹\mathbf{80}$

Similarly, $Z_4 = \min\{Z_0 + C_{1,4}\ , Z_1 + C_{2,4}\ , Z_2 + C_{3,4}\ , Z_3 + C_{4,4}\}$

Now, $C_{1,4} = \min\{C_{1,1,4}\ , C_{1,2,4}\ , C_{1,3,4}\ , C_{1,4,4}\}$

Optional

Calculation

$$C_{1,1,4} = A_1 + h_1(d_2 + d_3 + d_4) + h_2(d_3 + d_4) + h_3 d_4$$

$$C_{1,1,4} = 50 + (2)(40 + 20 + 10) + (2)(20 + 10) + (1)(10) = \textbf{₹260}$$

$$C_{1,2,4} = A_2 + h_2(d_3 + d_4) + h_3 d_4 + s_1 d_1$$

$$C_{1,2,4} = 20 + (2)(20 + 10) + (1)(10) + (3)(10) = \textbf{₹120}$$

$$C_{1,3,4} = A_3 + h_3 d_4 + s_1 d_1 + s_2(d_1 + d_2)$$

$$C_{1,3,4} = 30 + (1)(10) + (3)(10) + (3)(10 + 40) = \textbf{₹220}$$

$$C_{1,4,4} = A_4 + s_1 d_1 + s_2(d_1 + d_2) + s_3(d_1 + d_2 + d_3)$$

$$C_{1,4,4} = 50 + (3)(10) + (3)(10 + 40) + (3)(10 + 40 + 20) = \textbf{₹440}$$

Optional

Calculation

Therefore, $C_{1,4} = \min\{260, 120, 220, 440\} = ₹\mathbf{120}$

Now, $C_{2,4} = \min\{C_{2,2,4}\ , C_{2,3,4}\ , C_{2,4,4}\}$

$$C_{2,2,4} = A_2 + h_2(d_3 + d_4) + h_3 d_4 = 20 + (2)(20 + 10) + (1)(10) = ₹\mathbf{110}$$

$$C_{2,3,4} = A_3 + h_3 d_4 + s_2 d_2 = 30 + (1)(10) + (3)(40) = ₹\mathbf{160}$$

$$C_{2,4,4} = A_4 + s_2 d_2 + s_3(d_2 + d_3) = 50 + (3)(40) + (3)(40 + 20) = ₹\mathbf{350}$$

Therefore, $C_{2,4} = \min\{110, 160, 350\} = ₹\mathbf{110}$

Now, $C_{3,4} = \min\{C_{3,3,4}\ , C_{3,4,4}\}$

Optional

Calculation

$$C_{3,3,4} = A_3 + h_3 d_4 = 30 + (1)(10) = ₹\mathbf{40}$$

$$C_{3,4,4} = A_4 + s_3 d_3 = 50 + (3)(20) = ₹\mathbf{110}$$

Therefore, $C_{3,4} = \min\{40, 110\} = ₹\mathbf{40}$

Now, $C_{4,4} = C_{4,4,4} = A_4 = ₹\mathbf{50}$

$$\therefore\ Z_4 = \min\{0 + 120, 50 + 110, 50 + 40, 80 + 50\}$$

$$Z_4 = \min\{120, 160, 90, 130\} = ₹\mathbf{90}$$

Optional

Calculation

Summary of $C_{t,u,k}$ values:

t	k	$u = 1$	$u = 2$	$u = 3$	$u = 4$	$C_{t,k}$
1	1	**50**	-	-	-	50
	2	130	**50**	-	-	50
	3	210	**90**	210	-	90
	4	260	**120**	220	440	120
2	2	-	**20**	-	-	20
	3	-	**60**	150	-	60
	4	-	**110**	160	350	110
3	3	-	-	**30**	-	30
	4	-	-	**40**	110	40
4	4	-	-	-	**50**	50

Optional

Calculation

Summary of Z_k values:

k		$t = 1$	$t = 2$	$t = 3$	$t = 4$	Z_k
1	Z_{t-1}	0	-	-	-	**50**
	$C_{t,k}$	50	-	-	-	
	$Z_{t-1} + C_{t,k}$	**50**	-	-	-	
2	Z_{t-1}	0	50	-	-	**50**
	$C_{t,k}$	50	20	-	-	
	$Z_{t-1} + C_{t,k}$	**50**	70	-	-	
3	Z_{t-1}	0	50	50	-	**80**
	$C_{t,k}$	90	60	30	-	
	$Z_{t-1} + C_{t,k}$	90	110	**80**	-	
4	Z_{t-1}	0	50	50	80	**90**
	$C_{t,k}$	120	110	40	50	
	$Z_{t-1} + C_{t,k}$	120	160	**90**	130	

Optional

Calculation

- To find the optimal lot sizes, set $k = 4$ and find the period t at which the value of $Z_{t-1} + C_{t,4}$ is minimum, i.e., equal to Z_4.

- Here, the minimum cost occurs at $t = 3$. This implies that the last replenishment decision considers the demands in periods 3 and 4.

- Further, to determine the period (3 or 4) in which the order is to be placed, it is necessary to find the value of u ($t \leq u \leq k$) at which $C_{3,u,4}$ is minimum, i.e., equal to $C_{3,4}$.

$$C_{3,4} = \min\{40, 110\} = \text{₹}\mathbf{40}$$

Optional

Calculation

- Here, the minimum value of $C_{3,u,4}$ occurs at $u = 3$ and the ordering quantity will be the sum of demands in periods 3 and 4, i.e.,

$$Q_3 = d_3 + d_4 = 20 + 10 = \mathbf{30\ units}$$

- Next, set $k = 3 - 1 = 2$ and find the period t at which the value of $Z_{t-1} + C_{t,2}$ is minimum, i.e., equal to Z_2.

$$Z_2 = \min\{50, 70\} = \text{₹}\mathbf{50}$$

- Here, the minimum value occurs at $t = 1$. This implies that the second-last replenishment decision considers the demands in periods 1 and 2.

Optional

Calculation

- To determine the period (1 or 2) in which the order is to be placed, it is necessary to find the value of u ($t \leq u \leq k$) at which $C_{1,u,2}$ is minimum, i.e., equal to $C_{1,2}$.

$$C_{1,2} = \min\{130, 50\} = \text{₹}\mathbf{50}$$

- Here, the minimum value of $C_{1,u,2}$ occurs at $u = 2$ and the ordering quantity will be the sum of demands in periods 1 and 2, i.e.,

$$Q_2 = d_1 + d_2 = 10 + 40 = \mathbf{50}\ \boldsymbol{units}$$

Optional

Calculation

- Now, we have reached $t = 1$. This implies that the demands in all periods have been considered and the optimal lot sizes have been determined. Hence, the process terminates.

- The final solution is:

Period	1	2	3	4
Lot Size	-	50	30	-

- And the optimal cost is $Z_4 = ₹\mathbf{90}$.

Optional

Effect of Unit Price

- In the implementation of the W-W algorithm, we had not taken into account the unit purchase cost of the product while computing the total cost.

- If this unit price does not vary with time as well as ordering quantity (as implicitly assumed in the dynamic programming formulation) and no shortages or residual inventory are allowed at the end of the planning period, the total purchase cost will be the same (total demand x unit price) irrespective of the replenishment decisions made, and hence, it can be safely ignored while computing the total cost.

- However, if the unit price does vary with time or ordering quantity, the purchase cost must be included while computing $C_{t,k}$ or $C_{t,u,k}$, as the case may be.

Optional

Limitations of W-W Algorithm

- The complex nature of the algorithm makes it difficult for practitioners to understand and intuitively make sense of it.

- In worst-case scenarios, demand and cost information up to the last period may be needed just to calculate the initial replenishment quantity.

- The computational requirements of the algorithm rapidly increase when the number of periods increases, making it to difficult to scale to larger problems.

- It can only handle finite horizons, and so, if the actual planning horizon is infinite, it must be suitably approximated by rolling finite horizons before using the algorithm.

A General Dynamic Programming Model for Dynamic Lot Sizing

Let

T = No. of periods

L = Lead time

Z_k = State variable that represents the no. of units procured upto period (stage) k $(L + 1 \leq k \leq T)$

Z_{k-1} = Alternative with respect to period k $(L + 1 \leq k \leq T)$; $Z_{k-1} \leq Z_k$

P_k = No. of units procured in period k $(L + 1 \leq k \leq T)$; $P_k = Z_k - Z_{k-1}$

$F_k(Z_k)$ = Minimum total inventory cost, given Z_k units have been procured upto period k $(L + 1 \leq k \leq T)$

$C_k(Z_k, Z_{k-1})$ = Total inventory cost, given Z_k units have been procured upto period k and Z_{k-1} units have been procured upto period $k - 1$ $(L + 1 \leq k \leq T)$.

Optional

Optional

A General DP Model for Dynamic Lot Sizing

D_k = Demand in period k $(1 \leq k \leq T)$

A_k = Ordering cost per order in period k $(1 \leq k \leq T - L)$

h_k = Holding cost per unit in period k $(1 \leq k \leq T)$

s_k = Backorder cost per unit in period k $(1 \leq k \leq T)$

I_0 = Beginning Inventory Level

Y_k = Indicator function for order placement in period k $(1 \leq k \leq T - L)$

$$Y_k = \begin{cases} 1, & \text{if } Z_{k+L} > Z_{k+L-1} \\ 0, & \text{otherwise} \end{cases}$$

I_k = Inventory Level at the end of period k $(1 \leq k \leq T)$

$$I_k = \max\left\{0, Z_k + I_o - \sum_{t=1}^{k} D_t\right\}$$

B_k = Backorder quantity at the end of period k $(1 \leq k \leq T)$

$$B_k = \max\{0, \sum_{t=1}^{k} D_t - Z_k - I_o\}$$

Optional

A General DP Model for Dynamic Lot Sizing

- The dynamic programming forward recursive equation is written as:

$$F_k(Z_k) = \min_{Z_{k-1} \le Z_k \le Z_T} \{C_k(Z_k, Z_{k-1}) + F_{k-1}(Z_{k-1})\}$$

where $Z_k = 0 \; \forall \; k \in \{0,1,2, \dots, L\}$ and $F_0(Z_0) = 0$

- In order to ensure that no demand remains unfulfilled at the end of the planning horizon, the total procurement upto the last period T must be equal to the total demand minus any beginning inventory.

$$Z_T = \sum_{k=1}^{T} D_k - I_o$$

- The cost term $C_k(Z_k, Z_{k-1})$ is calculated as follows:

$$C_k(Z_k, Z_{k-1}) = A_{k-L} Y_{k-L} + h_k I_k + s_k B_k$$

Optional

Example

Period	Demand (units)	Ordering Cost (₹)	Holding Cost per unit (₹)	Shortage Cost per unit (₹)
1	20	50	2	3
2	40	20	2	3
3	20	30	1	3
4	10	50	1	3

Find the optimal lot sizes assuming an initial inventory level of 10 and zero lead time.

Optional

Solution

- Here, the period demands are multiples of 10. Hence, we consider each alternative in every stage only in multiples of 10, in order to reduce computational effort.

- Total quantity of items to be procured at the end of the last period $(T = 4)$ is given by:

$$Z_T = Z_4 = \sum_{k=1}^{4} D_k - I_0 = (20 + 40 + 20 + 10) - 10 = \mathbf{80\ units}$$

- Also, $Z_0 = 0$. The calculated values of $C_k(Z_k, Z_{k-1}) + F_{k-1}(Z_{k-1})$ from the execution of the DP model are tabulated in the subsequent slides.

Optional

Solution

$Z_1 \backslash Z_0$	0	$F_1(Z_1)$	Z_0*
0	30	30	0
10	50	50	0
20	70	70	0
30	90	90	0
40	110	110	0
50	130	130	0
60	150	150	0
70	170	170	0
80	190	190	0

Optional

Solution

$Z_2 \backslash Z_1$	0	10	20	30	40	50	60	70	80	$F_2(Z_2)$	Z_1*
0	180	-	-	-	-	-	-	-	-	180	0
10	170	170	-	-	-	-	-	-	-	170	0,10
20	140	160	160	-	-	-	-	-	-	140	0
30	110	130	150	150	-	-	-	-	-	110	0
40	80	100	120	140	140	-	-	-	-	80	0
50	50	70	90	110	130	130	-	-	-	50	0
60	70	90	110	130	150	170	170	-	-	70	0
70	90	110	130	150	170	190	210	210	-	90	0
80	110	130	150	170	190	210	230	250	250	110	0

Optional

Solution

$Z_3 \backslash Z_2$	0	10	20	30	40	50	60	70	80	$F_3(Z_3)$	Z_2*
0	390	-	-	-	-	-	-	-	-	390	0
10	390	350	-	-	-	-	-	-	-	350	10
20	360	350	290	-	-	-	-	-	-	290	20
30	330	320	290	230	-	-	-	-	-	230	30
40	300	290	260	230	170	-	-	-	-	170	40
50	270	260	230	200	170	110	-	-	-	110	50
60	240	230	200	170	140	110	100	-	-	100	60
70	210	200	170	140	110	80	100	90	-	80	50
80	220	210	180	150	120	90	110	130	120	90	**50**

Optional

Solution

$Z_4 \backslash Z_3$	0	10	20	30	40	50	60	70	80	$F_4(Z_4)$	Z_3*
80	440	400	340	280	220	160	150	130	90	90	80

- To find the optimal lot sizes, start with the last table and find where the minimum cost occurs, i.e., the value of Z_3^* corresponding to $Z_4 = 80$. Here, $\boldsymbol{Z_3^* = 80}$.

- Likewise, go to the previous table and find the value of Z_2^* corresponding to $Z_3 = 80$. Here, $\boldsymbol{Z_2^* = 50}$. By extending this procedure, we find $\boldsymbol{Z_1^* = 0}$.

- The ordering quantity (P_k) in each period k is given by the difference between Z_k and Z_{k-1}.

k	1	2	3	4
P_k	0	**50**	**30**	0

- The minimum cost is $\boldsymbol{F_4(Z_4) = ₹90}$

A General MILP for Dynamic Lot Sizing

Let

P = No. of periods;

L = Lead Time;

M = A large constant;

D_t = Demand in period t; $t \in \{1,2,3, \dots, P\}$

A_t = Ordering cost per order in period t; $t \in \{1,2,3, \dots, P - L\}$

H_t = Holding cost per unit in period t; $t \in \{1,2,3, \dots, P\}$

S_t = Backorder (Shortage) cost per unit in period t; $t \in \{1,2,3, \dots, P\}$

A General MILP for Dynamic Lot Sizing

Decision Variables:

Q_t = Quantity received at the beginning of period t for the order placed at the beginning of period $t - L$; $t \in \{L + 1,\ L + 2,\ L + 3, \dots, P\}$

(Q_t = 0 $\forall\ t \in \{1,2,3, \dots, L\}$)

I_t = Inventory Level at the end of period t; $t \in \{1,2,3, \dots, P\}$

B_t = Backorder Quantity at the end of period t; $t \in \{1,2,3, \dots, P\}$

$$Y_t = \begin{cases} 1, & if\ an\ order\ is\ placed\ in\ period\ t \\ 0, & otherwise \end{cases} ; t \in \{1,2,3, \dots, P - L\}$$

A General MILP for Dynamic Lot Sizing

Objective:

$$Min\ Z = \sum_{t=1}^{P-L} A_t Y_t + \sum_{t=1}^{P} H_t I_t + \sum_{t=1}^{P} S_t B_t$$

$subject\ to$

$$Q_t \leq M Y_{t-L}; \quad \forall\ t \in \{L+1,\ L+2,\ L+3, \dots, P\}$$

$$Q_t = 0; \quad \forall\ t \in \{1,2,3, \dots, L\}$$

$$I_t \geq \sum_{t'=1}^{t} (Q_{t'} - D_{t'}); \quad \forall\ t \in \{1,2,3, \dots, P\}$$

$$B_t \geq \sum_{t'=1}^{t} (D_{t'} - Q_{t'}); \quad \forall\ t \in \{1,2,3, \dots, P\}$$

$$I_P = 0;\ B_P = 0;$$

All Q_t, I_t and $B_t \geq 0$;

$$Y_t \in \{0,1\};\ \forall\ t \in \{1,2,3, \dots, P-L\}$$

Example

Using the same example as in the basic W-W algorithm,

No. of periods, $P = 10$; Lead time, $L = 0$

Ordering cost per order, $A =$ ₹300 (for all periods)

Holding cost per unit, $H =$ ₹1 (for all periods)

Backordering is not permitted here.

Demand in each period (D_t) given in following table:

Period t	1	2	3	4	5	6	7	8	9	10
d_t	50	60	90	70	30	100	60	40	80	20

Since the total quantity to be ordered across all periods cannot exceed the total demand in an optimal solution, the Big-M value may be approximated by the total demand (600 in this case).

Solution

- Since backordering is not permitted here, two approaches can be followed:

Approach 1:

- Set unit backorder cost to an extremely large value ($S_t \to \infty, \forall\ t$) and use the same general MILP model as described previously.

Approach 2:

- Add an additional constraint $B_t = 0\ \forall\ t$, to the general MILP model that prohibits backordering and forces demand satisfaction in every period.

Solution following Approach 1

An optimal MILP solution:

t	1	2	3	4	5	6	7	8	9	10
Q	110	0	190	0	0	300	0	0	0	0
I	60	0	100	30	0	200	140	100	20	0
B	0	0	0	0	0	0	0	0	0	0

The optimal cost is ₹$\mathbf{1550}$.

Modified MILP for zero backlogging (Approach 2)

Objective:

$$Min\ Z = \sum_{t=1}^{P-L} A_t Y_t + \sum_{t=1}^{P} H_t I_t + \sum_{t=1}^{P} S_t B_t$$

$subject\ to$

$$Q_t \leq M Y_{t-L}; \quad \forall\ t \in \{L+1,\ L+2,\ L+3, \dots, P\}$$

$$Q_t = 0; \quad \forall\ t \in \{1,2,3, \dots, L\}$$

$$I_t \geq \sum_{t'=1}^{t} (Q_{t'} - D_{t'}); \quad \forall\ t \in \{1,2,3, \dots, P\}$$

$$B_t \geq \sum_{t'=1}^{t} (D_{t'} - Q_{t'}); \quad \forall\ t \in \{1,2,3, \dots, P\}$$

$$\boldsymbol{B_t = 0}; \quad \forall\ \boldsymbol{t} \in \{\mathbf{1}, \mathbf{2}, \mathbf{3}, \dots, \boldsymbol{P}\}$$

$$I_P = 0;$$

All Q_t and $I_t \geq 0$;

$$Y_t \in \{0,1\};\ \forall\ t \in \{1,2,3, \dots, P-L\}$$

Solution following Approach 2

An optimal MILP solution:

t	1	2	3	4	5	6	7	8	9	10
Q	110	0	190	0	0	300	0	0	0	0
I	60	0	100	30	0	200	140	100	20	0
B	0	0	0	0	0	0	0	0	0	0

The optimal cost is ₹$\mathbf{1550}$. We find that both approaches yield the same results.

Optional

LAGRANGIAN RELAXATION (LR)

- In order to efficiently find good solutions to hard combinatorial optimization problems via a branch and bound implementation, the following need to be considered:
  - calculation of a lower bound that is as close as possible to the optimum, and
  - calculation of an upper bound that is as close as possible to the optimum.

- In case of a minimization (maximization) problem, a lower (upper) bound represents the objective value of an optimal solution to a relaxed version of the original problem, while an upper (lower) bound represents the best known feasible solution to the original problem.

Optional

LAGRANGIAN RELAXATION (LR)

- Here, ‘relaxed’ refers to a formulation of the problem with fewer constraints that is easier to solve to optimality compared to the original problem formulation.

- Since a maximization problem can be reformulated as a minimization problem by simply negating the objective function, (i.e., If $y = f(x)$ is the objective function, maximizing y is equivalent to minimizing $-y$) we henceforth consider things in this discussion only from the perspective of a minimization problem.

Optional

LAGRANGIAN RELAXATION (LR)

- To find a good initial upper bound, a heuristic procedure may be used. This upper bound can be further updated if better feasible solutions are found during the branch and bound implementation.

- To find lower bounds, one well-known general technique is **Linear Programming (LP) relaxation**. In LP relaxation we take an integer (or mixed-integer) programming formulation of the problem and relax the integrality requirement on the variables.

- Another well-known (and well-used) technique which is available to find lower bounds is **Lagrangian relaxation (LR)**.

Optional

Lagrangian relaxation involves:

1) taking an integer (or mixed-integer) programming formulation of the problem;

2) attaching Lagrange multipliers (λ) to some of the constraints in this formulation and relaxing these constraints into the objective function;

3) solving (exactly) the resulting integer (or mixed-integer) program.

The value of :

$$\begin{array}{ll} \text{minimize} & cx \\ \textit{subject to} & \\ & Ax \geq b \\ & Bx \geq d \\ & x \geq 0 \end{array} \quad \geq \quad \begin{array}{ll} \text{minimize} & cx + \lambda(b - Ax) \\ \textit{subject to} & Ax \geq b \\ & Bx \geq d \\ & x \geq 0 \end{array}$$

(Since $\lambda \geq 0$ and $(b - Ax) \leq 0$, we are merely adding a non-positive term to the objective function and hence, its value cannot increase.)

Optional

This in turn is greater than or equal to the value of:

$$\text{minimize} \quad cx + \lambda(b - Ax)$$

$$subject\ to$$

$$Bx \geq d$$

$$x \geq 0$$

(This is because removing a set of constraints from a minimization problem will not lead to an increase in the objective function value.)

The program after Lagrangian relaxation is:

$$\text{minimize} \quad cx + \lambda(b - Ax) = (\mathrm{c} - \lambda A)x + \lambda b$$

$$subject\ to$$

$$Bx \geq d$$

$$x \geq 0$$

Optional

LR in Inventory

- Consider the general MILP model for dynamic lot sizing introduced in the previous slides.

- In that model, it was implicitly assumed that there is no restriction on the quantity of items that can be ordered in any given period. What if this is no longer the case?

- Similarly, what happens when there is a capacity restriction on the number of units that can be held in the inventory at any point of time? What if, for instance, the management does not want the total procurement cost to exceed a certain budget limit?

Optional

LR in Inventory

- Each of the these scenarios can be handled by including suitable additional constraints into the general MILP model, at the cost of making it *harder* to solve the model.

- LR provides an elegant and computationally efficient way of incorporating these constraints into the model without explicitly stating them.

- In the LR technique, instead of directly including the 'hard' constraints into the formulation, the negative of the slack of each 'hard' constraint is scaled by a multiplier and added to the objective function.

Optional

LR in Inventory

- The optimal solution to the relaxed problem provides a *lower bound* on the optimal solution to the original problem, i.e., the optimal objective value of the original problem cannot be lower than that of the relaxed problem.

- In theory, finding the best value for the penalty multiplier involves exactly solving the *Lagrangian dual* of the problem, which is another hard problem.

- In practice, however, approximate iterative methods like the sub-gradient method may be used to estimate the multiplier.

Optional

LR-Based MILP for Budget-Constrained Multi-Item Inventory System

- Consider a multi-item inventory system in which the objective is to minimize the total inventory costs, with a restriction imposed on the total procurement (ordering and purchase) expenditure over the planning period.

- Let there be n items, U_{it} be the unit cost of the ith item in period t and C be the allowed budget limit. For simplicity, let the lead time be zero and consider P periods.

- The MILP and the LR-based MILP model are provided in the two subsequent slides:

Optional

MILP:

$$Min\ Z = \sum_{i=1}^{n}\sum_{t=1}^{P} A_{it}Y_{it} + \sum_{i=1}^{n}\sum_{t=1}^{P} H_{it}I_{it} + \sum_{i=1}^{n}\sum_{t=1}^{P} S_{it}B_{it} + \sum_{i=1}^{n}\sum_{t=1}^{P} U_{it}Q_{it}$$

subject to

$$Q_{it} \leq MY_{it}; \quad \forall\, i \in \{1,2,3,\dots,n\}, t \in \{1,2,3,\dots,P\}$$

$$I_{it} \geq \sum_{t'=1}^{t} (Q_{it'} - D_{it'})\,; \quad \forall\, i \in \{1,2,3,\dots,n\}, t \in \{1,2,3,\dots,P\}$$

$$B_{it} \geq \sum_{t'=1}^{t} (D_{it'} - Q_{it'})\,; \quad \forall\, i \in \{1,2,3,\dots,n\}, t \in \{1,2,3,\dots,P\}$$

$$\boldsymbol{\sum_{i=1}^{n}\sum_{t=1}^{P} (A_{it}Y_{it} + U_{it}Q_{it}) \leq C}$$

$$I_{iP} = 0\,; \quad \forall\, i \in \{1,2,3,\dots,n\}$$

$$B_{iP} = 0\,; \quad \forall\, i \in \{1,2,3,\dots,n\}$$

All Q_{it}, I_{it} and $B_{it} \geq 0$;

$$Y_{it} \in \{0,1\};\ \forall\, i \in \{1,2,3,\dots,n\}, t \in \{1,2,3,\dots,P\}$$

LR-Based MILP: Optional

$$Min\ Z_{LR} = \sum_{i=1}^{n}\sum_{t=1}^{P} A_{it}Y_{it} + \sum_{i=1}^{n}\sum_{t=1}^{P} H_{it}I_{it} + \sum_{i=1}^{n}\sum_{t=1}^{P} S_{it}B_{it} + \sum_{i=1}^{n}\sum_{t=1}^{P} U_{it}Q_{it} + \boldsymbol{\lambda W}$$

subject to

$$Q_{it} \leq MY_{it}; \quad \forall\, i \in \{1,2,3,\dots,n\}, t \in \{1,2,3,\dots,P\}$$

$$I_{it} \geq \sum_{t'=1}^{t} (Q_{it'} - D_{it'})\,; \quad \forall\, i \in \{1,2,3,\dots,n\}, t \in \{1,2,3,\dots,P\}$$

$$B_{it} \geq \sum_{t'=1}^{t} (D_{it'} - Q_{it'})\,; \quad \forall\, i \in \{1,2,3,\dots,n\}, t \in \{1,2,3,\dots,P\}$$

$$\boldsymbol{W \geq \sum_{i=1}^{n}\sum_{t=1}^{P} (A_{it}Y_{it} + U_{it}Q_{it}) - C}$$

$$I_{iP} = 0\,; \quad \forall\, i \in \{1,2,3,\dots,n\}$$

$$B_{iP} = 0\,; \quad \forall\, i \in \{1,2,3,\dots,n\}$$

All Q_{it}, I_{it}, B_{it} and $W \geq 0$;

$$Y_{it} \in \{0,1\};\ \forall\, i \in \{1,2,3,\dots,n\}, t \in \{1,2,3,\dots,P\}$$

Here $\lambda > 0$ is the Lagrangian multiplier

Optional

Example

Item	Period	Demand (units)	Ordering Cost (₹)	Holding Cost per unit (₹)	Shortage Cost per unit (₹)	Unit Price (₹)
1	1	25	50	1	1	2
	2	40	40	1	2	2
	3	20	50	1	3	1
2	1	30	20	2	2	2
	2	20	20	3	3	1
	3	45	25	3	5	3
3	1	15	30	1	3	1
	2	40	30	2	3	2
	3	35	20	2	4	2

- Find the optimal lot sizes for each item if the total procurement costs across the three periods must not exceed ₹500.

Optional

Solution

- The optimal lot sizes for each item at an estimated multiplier value of $\lambda = 0.2$ are as follows:

Item	Period	Lot size, Q (units)	Inventory Level, I (units)	Backorder Quantity, B (units)	Ordering Cost incurred	Holding Cost incurred	Shortage Cost incurred	Purchase Cost incurred
1	1	-	0	25	-	-	25	-
	2	85	20	0	40	20	-	170
	3	-	0	0	-	-	-	-
2	1	-	0	30	-	-	60	-
	2	95	45	0	20	135	-	95
	3	-	0	0	-	-	-	-
3	1	55	40	0	30	40	-	55
	2	-	0	0	-	-	-	-
	3	35	0	0	20	-	-	70
				Total	110	195	85	390

- The total cost is ₹$\mathbf{780}$.

ABC Analysis

- ABC analysis divides on-hand inventory into three classifications on the basis of dollar (TL) volume (usage value)
- It is based on Pareto Principle. (80-20 principle)
- The idea is to focus resources on the ***critical few*** and not on the ***trivial many***.

- ABC analysis divides the inventory into 3 categories.

Category A	Low in number High Usage Value Items
Category B	Medium in number and Medium Usage Value items
Category C	High in Number Low in Usage Value

One can take the example of a Furniture Store

Step 1**:** Multiply the total number of items by the cost of each unit to find the annual usage value.

Products	Annual Number of Items Sold	Costs per unit	Annual Usage Value
Beds	5000	$80	$400,000
Chairs	1500	$20	$30,000
Coffee Tables	700	$40	$28,000
Desks	600	$40	$30,000
Ottomans	500	$30	$15,000
Dining Table	700	$50	$35,000
Book Cases	600	$15	$9,000
Office Chairs	10,000	$20	$200,000
Wardrobes	600	$40	$24,000
Computer Cabinet	700	$30	$21,000

- Step 2: After noting all the products of the inventory, it’s time to list them in the descending order based on annual consumption value.

- Step 3: Sum up and add the total number of units sold and the annual consumption value.

- Step 4: Find out the **cumulative percentage of products sold** along with the percentage of annual consumption value.

Products	Annual Number of Items Sold	Costs per unit	Annual Usage Value	Percentage of Annual Units Sold	Percentage of Annual Consumption Value
Beds	5000	$80	$400,000	23.80	52.00
Office Chairs	10,000	$20	$200,000	47.61	27.00
Dining Table	700	$50	$35,000	3.33	5.06
Chairs	1500	$20	$30,000	7.14	4.15
Desks	700	$40	$28,000	3.33	4.03
Coffee Tables	600	$40	$24,000	2.85	2.08
Wardrobes	600	$20	$24,000	2.85	2.08
Computer Cabinet	700	$30	$21,000	3.33	2.02
Ottomans	500	$30	$15,000	2.38	1.05
Book Cases	600	$15	$9,000	2.85	1.01
Totel	**20,900**		**$771,000**		

Step 5: In the last step, split the data and numbers into the three A, B, and C categories. Remember, it's essential to set the data in the ratio of 80:15:5.

	Products	Annual Number of Items Sold	Costs per unit	Annual Usage Value	Percentage of Annual Units Sold	Percentage of Annual Consumption Value
79%	Beds	5000	$80	$400,000	23.80	52.00
	Office Chairs	10,000	$20	$200,000	47.61	27.00
13%	Dining Table	700	$50	$35,000	3.33	5.06
	Chairs	1500	$20	$30,000	7.14	4.15
	Desks	700	$40	$28,000	3.33	4.03
8%	Coffee Tables	600	$40	$24,000	2.85	2.08
	Wardrobes	600	$20	$24,000	2.85	2.08
	Computer Cabinet	700	$30	$21,000	3.33	2.02
	Ottomans	500	$30	$15,000	2.38	1.05
	Book Cases	600	$15	$9,000	2.85	1.01
	Totel	**20,900**		**$771,000**		

A category

B category

C category

Thank You !!!